MITSUBISHI ELECTRIC

Mitsubishi Electric Industrial Robot

CR800 Series Controller

INSTRUCTION MANUAL

Troubleshooting

This instruction manual apply to both the CR800-R/CR800-Q controller corresponding to iQ Platform, and the CR800-D controller of standalone type.

MELFA
BFP-A3480-F

⚠ Safety Precautions

Always read the following precautions and the separate "Safety Manual" before starting use of the robot to learn the required measures to be taken.

⚠ CAUTION

All teaching work must be carried out by an operator who has received special training. (This also applies to maintenance work with the power source turned ON.)
Enforcement of safety training

⚠ CAUTION

For teaching work, prepare a work plan related to the methods and procedures of operating the robot, and to the measures to be taken when an error occurs or when restarting. Carry out work following this plan. (This also applies to maintenance work with the power source turned ON.)
Preparation of work plan

⚠ WARNING

Prepare a device that allows operation to be stopped immediately during teaching work. (This also applies to maintenance work with the power source turned ON.)
Setting of emergency stop switch

⚠ CAUTION

During teaching work, place a sign indicating that teaching work is in progress on the start switch, etc. (This also applies to maintenance work with the power source turned ON.)
Indication of teaching work in progress

⚠ DANGER

Provide a fence or enclosure during operation to prevent contact of the operator and robot.
Installation of safety fence

⚠ CAUTION

Establish a set signaling method to the related operators for starting work, and follow this method.
Signaling of operation start

⚠ CAUTION

As a principle turn the power OFF during maintenance work. Place a sign indicating that maintenance work is in progress on the start switch, etc.
Indication of maintenance work in progress

⚠ CAUTION

Before starting work, inspect the robot, emergency stop switch and other related devices, etc., and confirm that there are no errors.
Inspection before starting work

The points of the precautions given in the separate "Safety Manual" are given below.
Refer to the actual "Safety Manual" for details.

⚠ DANGER	When automatic operation of the robot is performed using multiple control devices (GOT, programmable controller, push-button switch), the interlocking of operation rights of the devices, etc. must be designed by the customer.
⚠ CAUTION	Use the robot within the environment given in the specifications. Failure to do so could lead to a drop or reliability or faults. (Temperature, humidity, atmosphere, noise environment, etc.)
⚠ CAUTION	Transport the robot with the designated transportation posture. Transporting the robot in a non-designated posture could lead to personal injuries or faults from dropping.
⚠ CAUTION	Always use the robot installed on a secure table. Use in an instable posture could lead to positional deviation and vibration.
⚠ CAUTION	Wire the cable as far away from noise sources as possible. If placed near a noise source, positional deviation or malfunction could occur.
⚠ CAUTION	Do not apply excessive force on the connector or excessively bend the cable. Failure to observe this could lead to contact defects or wire breakage.
⚠ CAUTION	Make sure that the workpiece weight, including the hand, does not exceed the rated load or tolerable torque. Exceeding these values could lead to alarms or faults.
⚠ WARNING	Securely install the hand and tool, and securely grasp the workpiece. Failure to observe this could lead to personal injuries or damage if the object comes off or flies off during operation.
⚠ WARNING	Securely ground the robot and controller. Failure to observe this could lead to malfunctioning by noise or to electric shock accidents.
⚠ CAUTION	Indicate the operation state during robot operation. Failure to indicate the state could lead to operators approaching the robot or to incorrect operation.
⚠ WARNING	When carrying out teaching work in the robot's movement range, always secure the priority right for the robot control. Failure to observe this could lead to personal injuries or damage if the robot is started with external commands.
⚠ CAUTION	Keep the jog speed as low as possible, and always watch the robot. Failure to do so could lead to interference with the workpiece or peripheral devices.
⚠ CAUTION	After editing the program, always confirm the operation with step operation before starting automatic operation. Failure to do so could lead to interference with peripheral devices because of programming mistakes, etc.
⚠ CAUTION	Make sure that if the safety fence entrance door is opened during automatic operation, the door is locked or that the robot will automatically stop. Failure to do so could lead to personal injuries.
⚠ CAUTION	Never carry out modifications based on personal judgments, or use non-designated maintenance parts. Failure to observe this could lead to faults or failures.

⚠ WARNING

When the robot arm has to be moved by hand from an external area, do not place hands or fingers in the openings. Failure to observe this could lead to hands or fingers catching depending on the posture.

⚠ CAUTION

Do not stop the robot or apply emergency stop by turning the robot controller's main power OFF. If the robot controller main power is turned OFF during automatic operation, the robot accuracy could be adversely affected. Moreover, it may interfere with the peripheral device by drop or move by inertia of the arm.

⚠ CAUTION

Do not turn off the main power to the robot controller while rewriting the internal information of the robot controller such as the program or parameters. If the main power to the robot controller is turned off while in automatic operation or rewriting the program or parameters, the internal information of the robot controller may be damaged.

⚠ DANGER

Do not connect the Handy GOT when using the GOT direct connection function of this product. Failure to observe this may result in property damage or bodily injury because the Handy GOT can automatically operate the robot regardless of whether the operation rights are enabled or not.

⚠ DANGER

Do not connect the Handy GOT to a programmable controller when using an iQ Platform compatible product with the CR800-R/CR800-Q controller. Failure to observe this may result in property damage or bodily injury because the Handy GOT can automatically operate the robot regardless of whether the operation rights are enabled or not.

⚠ DANGER

Do not remove the SSCNET III cable while power is supplied to the multiple CPU system or the servo amplifier. Do not look directly at light emitted from the tip of SSCNET III connectors or SSCNET III cables of the Motion CPU or the servo amplifier. Eye discomfort may be felt if exposed to the light. (Reference: SSCNET III employs a Class 1 or equivalent light source as specified in JIS C 6802 and IEC60825-1 (domestic standards in Japan).)

⚠ DANGER

Do not remove the SSCNET III cable while power is supplied to the controller. Do not look directly at light emitted from the tip of SSCNET III connectors or SSCNET III cables. Eye discomfort may be felt if exposed to the light. (Reference: SSCNET III employs a Class 1 or equivalent light source as specified in JIS C 6802 and IEC60825-1 (domestic standards in Japan).)

⚠ DANGER

Attach the cap to the SSCNET III connector after disconnecting the SSCNET III cable. If the cap is not attached, dirt or dust may adhere to the connector pins, resulting in deterioration connector properties, and leading to malfunction.

⚠ CAUTION

Make sure there are no mistakes in the wiring. Connecting differently to the way specified in the manual can result in errors, such as the emergency stop not being released. In order to prevent errors occurring, please be sure to check that all functions (such as the teaching box emergency stop, customer emergency stop, and door switch) are working properly after the wiring setup is completed.

⚠ CAUTION

Use the network equipments (personal computer, USB hub, LAN hub, etc) confirmed by manufacturer. The thing unsuitable for the FA environment (related with conformity, temperature or noise) exists in the equipments connected to USB. When using network equipment, measures against the noise, such as measures against EMI and the addition of the ferrite core, may be necessary. Please fully confirm the operation by customer. Guarantee and maintenance of the equipment on the market (usual office automation equipment) cannot be performed.

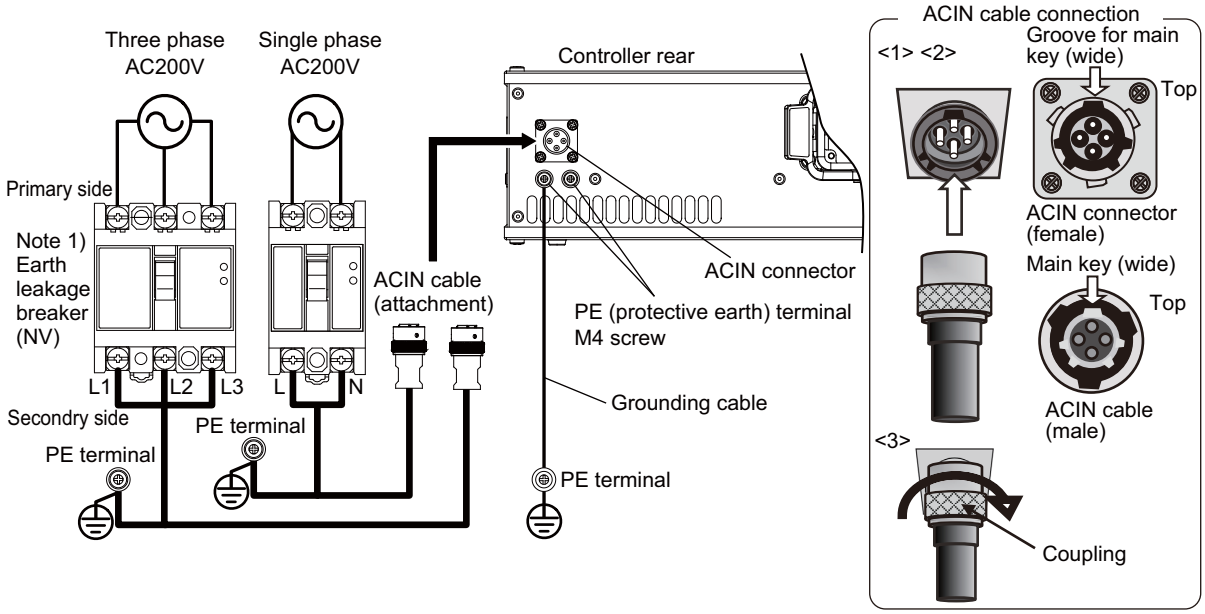
⚠ CAUTION

To maintain the safety of the robot system against unauthorized access from external devices via the network, take appropriate measures.
To maintain the safety against unauthorized access via the Internet, take measures such as installing a firewall.

*CR800 controller

Notes of the basic component are shown.

⚠ CAUTION Please install the earth leakage breaker in the primary side power supply of the controller because of leakage protection.



Note 1) Always use the terminal cover for the earth leakage breaker.

1) Prepare the following items.

Part name	Specifications	Remarks
Earth leakage breaker	The following is recommended product. Single phase: NV30FAU-2P-10A-AC100-240V-30mA (Terminal cover: TCS-05FA2) Three phase: NV30FAU-3P-10A-AC100-240V-30mA (Terminal cover: TCS-05FA3)	Prepared by customer.
Cable for primary power supply	AWG14 ($2mm^2$) or above	Prepared by customer. Tightening torque for terminal fixing screw is 2 to 3Nm.
Grounding cable	AWG14 ($2mm^2$) or above	Prepared by customer. Tightening torque for terminal fixing screw is 2 to 3Nm.
ACIN cable	Terminal: M5, cable length: 3m	Supplied with the product.

2) Confirm that the primary power matches the specifications.
3) Confirm that the primary power is OFF and that the earth leakage breaker power switch is OFF.
4) Connect the ACIN cable to the breaker.
   Connect the power terminals of the ACIN cable to the secondary side terminals of the earth leakage breaker. Also, ground the FG terminal of the cable.
5) Connect the ACIN cable to the ACIN connector on the rear of the controller.
   <1> Face the main key on the ACIN cable plug upwards. (Refer to the "ACIN cable connection" illustration.)
   <2> Align the main key of the ACIN cable plug with the grooves on the ACIN connector. Push the plug into the connector as far as it will go.
   The plug may be damaged if it is not correctly aligned with the connector.
   <3> Tighten the coupling on the ACIN cable, turning it to the right until it locks.
6) Connect one end of the grounding cable to the PE (protective earth) terminal on the controller and ground the other end (2-point grounding) in order to comply with the requirements of EN 61800-5-1 for the touch current of 3.5 mA AC or more.
7) Connect the primary power cable to the primary side terminal of the earth leakage breaker.

Revision history

Date of print	Specifications No.	Details of revisions
2017-05-25	BFP-A3480	・First print
2017-06-27	BFP-A3480-A	・Errors were added. H0315, H0316, H0317, L2660, L2661, L2662, H2663, L6011, H7860, H8740, H8741, H8751, H8752, H8760, H8761, H8770, H8780, H8790 ・Measures of error H887n were changed.
2017-09-25	BFP-A3480-B	・Supplementary explanation were added to measures of error C0152 and H0083.
2018-02-01	BFP-A3480-C	・Descriptions about the CR800-Q controller were added.
2018-09-01	BFP-A3480-D	・Supplementary explanation were added to measures of error C0152.
2018-10-15	BFP-A3480-E	・Supplementary explanation were added to measures of error L2611.
2018-12-25	BFP-A3480-F	・Added further explanation of the ACIN cable.

■ Introduction

Thank you for purchasing the Mitsubishi industrial robot. This instruction manual describes the causes and measures for errors that may occur while using the robot.
If an error should occur, refer to this manual and take appropriate measures.
Apply to both the CR800-R/CR800-Q series controller corresponding to iQ Platform, and the CR800-D series controller of standalone.

Notice

*ONLY QUALIFIED SERVICE PERSONNEL MAY INSTALL OR SERVICE THE ROBOT SYSTEM.
*ANY PERSON WHO PROGRAM, TEACHES, OPERATE, MAINTENANCE OR REPAIRS THE ROBOT SYSTEM IS TRAINED AND DEMONSTRATES COMPETENCE TO SAFELY PERFORM THE ASSIGNED TASK.
*ENSURE COMPLIANCE WITH ALL LOCAL AND NATIONAL SAFETY AND ELECTRICAL CODES FOR THE INSTALLATION AND OPERATION OF THE ROBOT SYSTEM.

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・The information contained in this document has been written to be accurate as much as possible. Please interpret that items not described in this document "cannot be performed." or "alarm may occur".
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Contents

Page

1 Error list 1-1
(1) Error No. 1-1
(2) Cause and measures against the error 1-1

2 Appendix Appendix-64
2.1 Place where fuse replacement is required Appendix-64
2.1.1 Place where a brake fuse replacement is required. Appendix-64
2.1.2 Place where hand fuse replacement is required Appendix-65
2.1.3 Place where fuse replacement is required. Appendix-66
2.1.4 Place where a converter fuse replacement is required Appendix-67
2.2 Fan installation place of robot controller. Appendix-68
2.3 Troubles and measures Appendix-69
2.4 Errors involving change in specification Appendix-72
2.5 Force sense interface unit errors Appendix-73

1 Error list

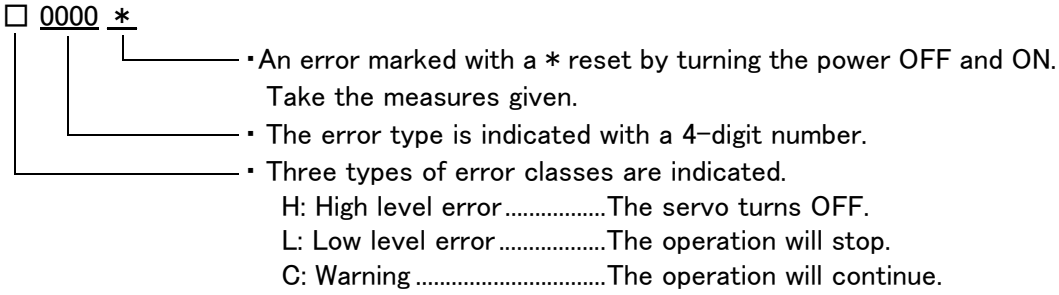
(1) Error No.

When an error occurs, ERROR LED at the front of the controller will turn on or blink.

ERROR LED status	Details
On	Low-level error, or warning occurred.
Flashing	High-level error occurred.
Off	Normal operation.

The four-digit error number (number except the one character of the head.) is displayed on LCD of T/B.
Example: In the case of C0010, display the display and the error message for "0010."
In addition, the alarm rings at 0.5-second intervals while an error is occurring. When resetting the power supply, if the interval between turn-on and turn-off of a controller is too short, the alarm rings at 0.1-second intervals.
The message, cause and measures to be taken are displayed in Table 1-1 for the error Nos. that may appear.
Also, a detailed message will be displayed on the Error History screen of the T/B, depending on the error No. of the error occurred. Check by displaying the Error History screen after resetting the error.
If the error recurs even after the measures in the table are taken, contact your service provider.

[Note] ・ The meaning of the error Nos. in Table 1-1 are shown below.



・ The axis No. may be indicated at the last digit of the error No.
Example) H0931 No. 1 axis motor overcurrent.

(2) Cause and measures against the error

The details, cause and measures of the error number occurrence are shown in Table 1-1.
Note) The contents of the error caused with option products are written in the instruction manual of the option. Refer to each instruction manual.

Table 1-1 : Error list

Error No.	Error cause and measures	
H0001	Error message	Fail safe error (SRVOFF)
	Cause	The system may be abnormal.
	Measures	Turn the power OFF and ON once. If it comes back, contact to your service provider.
H0002	Error message	Fail safe error (STOP)
	Cause	The system may be abnormal.
	Measures	Turn the power OFF and ON once. If it comes back, contact to your service provider.
H0003	Error message	The system is abnormal.
	Cause	The problem of the system is the cause.
	Measures	If it comes back, contact to your service provider.
H0004 ＊	Error message	CPU Watch dog error
	Cause	CPU was not normally treatable
	Measures	It is necessary to change some parts when not improvement. If it comes back, contact to your service provider. ＊ The "CPU" part in each message varies according to the CPU in which an error is detected. RCPU_main, CCPU_main, RCPU_sub, CCPU_sub, CPU_main, CPU_sub, FPGA, RCPU_PLD, CCPU_PLD

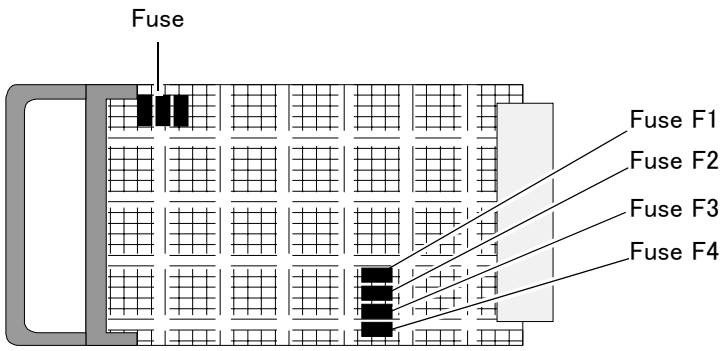
Error No.	Error cause and measures	
H0006 *	Error message	CPU process error
	Cause	CPU was not normally treatable
	Measures	It is necessary to change some parts when not improvement. * The "CPU" part in each message varies according to the CPU in which an error is detected. RCPU_main, CCPU_main, RCPU_sub, CCPU_sub, CPU_main, CPU_sub, FPGA
H0008 *	Error message	CPU synchronization error
	Cause	CPU synchronization was failed
	Measures	Turn the power OFF and ON once.
H0009 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Version UP (ALL)
	Cause	Version UP (ALL)
	Measures	Turn the power OFF and power ON once
	Error message	Version UP (MAIN)
	Cause	Version UP (MAIN)
	Measures	Turn the power OFF and power ON once
	Error message	Version UP (SERVO)
	Cause	Version UP (SERVO)
	Measures	Turn the power OFF and power ON once
	Error message	The servo s/w was written
	Cause	The servo s/w was written
	Measures	Please release an error by reset operation * This error can be reset without turning on the power supply again.
	Error message	Rewriting failed servo(Er.*)
	Cause	Servo S/W couldn't be read.
	Measures	When it occurs, contact your service provider * The number (1 to 7) which indicates the cause of the error is shown in "*" in the error message.
C0010	Error message	Illegal Version (file)
	Cause	The version is inconsistent.
	Measures	The file has been automatically initialized. The program is being deleted.
C0011	Error message	Illeagal Version (system data)
	Cause	The version is inconsistent.
	Measures	The file has been automatically initialized. Turn the power OFF and ON once.
C0012	Error message	Initialize (error log)
	Cause	The error log has been initialized because of version mismatch or the error log file is abnormal.
	Measures	Reset the alarm, and continue the operation.
C0013 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Illeagal file
	Cause	Data including programs may have been damaged.
	Measures	Contact your service provider as the initialization operation is required.
	Error message	Illegal system SD card
	Cause	File Crash
	Measures	File Initial operation
H0014 *	Error message	System error (illegal MECHA)
	Cause	A character string cannot exceed 14 characters.
	Measures	Re-input the correct name.
H0015 *	Error message	Illeagal Version (file)
	Cause	Illeagal Version (file)
	Measures	Contact the maker.

Error No.	Error cause and measures	
L0016 *	Error message	Turn the power OFF and ON once
	Cause	The time from turning the power OFF to turning the power ON again is too short.
	Measures	Give more time before turning the power ON again after turning the power OFF.
C0018	Error message	The robot model was selected.
	Cause	The robot model was selected.
	Measures	Reset the error.
H0020 *	Error message	System Error (same name is Backup data.)
	Cause	The data of the system backup area is abnormal.
	Measures	Please consult your service provider.
H0021 *	Error message	System Error (Backup data is Count over.)
	Cause	The control region is overflowing.
	Measures	Please consult your service provider.
H0022 *	Error message	System Error (Backup data is no area.)
	Cause	The region is too small.
	Measures	Please consult your service provider.
H0028 *	Error message	Memory error (ECC)
	Cause	2bit error of ECC was detected (***)
	Measures	Turn the power OFF and ON once * The part where an error is detected is shown in "***" in the message. nvSRAM, FROM, RCPU_DRAM, RCPU_L2C, CCPU_DRAM, CCPU_L2C, R_nvSRAM, R_DRAM, R_L2C
C0029	Error message	Memory error (ECC)
	Cause	1bit error of ECC was detected (***)
	Measures	When it frequently occurs, contact your service provider * The part where an error is detected is shown in "***" in the message. nvSRAM, FROM, RCPU_DRAM, RCPU_L2C, CCPU_DRAM, CCPU_L2C, R_nvSRAM, R_DRAM, R_L2C
L0030	Error message	Hand error. LS release
	Cause	This is a user setting error.
	Measures	Reset the error after removing the cause.
L0031	Error message	Air pressure error
	Cause	This is a user setting error.
	Measures	Reset the error after releasing the cause.
C0032	Error message	HIOTYPE parameter not set
	Cause	HIOTYPE parameter needs to be changed
	Measures	Please set the HIOTYPE parameter (-1: Not set/0: Source/1: Sink)
H0039	Error message	Door Switch Signal line is faulty.
	Cause	The one point of contact in 2 points of contact of the door switch has broken. Or wiring is not the double lines.
	Measures	Turn off the power supply. Confirm whether there is any problem in wiring of the switch. And, please confirm whether it is wiring of the double line. Refer to the "Examples of safety measures" given in separate "Standard Specifications Manual" for door switch wiring. Turn on the power supply again after checking.
H0040	Error message	Door Switch Signal is Input
	Cause	The door switch is open.
	Measures	Confirm whether the door switch input signal is connected correctly. And close the door connected to the input signal of door switch.

Error No.	Error cause and measures	
H0041 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Comm. error (Remote I/O #1).
	Cause	Communication line is illegal.
	Measures	In CR800-R/CR800-Q controller, confirm that the remote I/O cable between the CPU and the controller is grounded and connected correctly.
	Error message	The CRC error of remote I/O channel 1 occurs
	Cause	An error was found in the communication line for remote I/O channel 1.
	Measures	In CR800-R/CR800-Q controller, confirm that the remote I/O cable between the CPU and the controller is grounded and connected correctly.
H0042 *	Error message	Comm. error (Remote I/O #2).
	Cause	Communication line is illegal.
	Measures	Checks the remote I/O cable connection in the CR750-Q/CR751-Q controller.
H0044 *	Error message	Comm. error (Remote I/O #4)
	Cause	Communication line is illegal.
	Measures	Check the cable or power supply
H0045 *	Error message	Comm. error (Remote I/O #5)
	Cause	Communication line is illegal.
	Measures	Check the cable or power supply
H0046	Error message	Faulty wiring (Mode sel. switch).
	Cause	The state of doubled wiring is not matching (Mode sel. switch).
	Measures	Turn off the power and confirm whether wiring of the switch is right. Wiring needs to be doubled. Refer to the separate manual, "Standard Specifications Manual" for wiring of the mode selector switch.
H0047	Error message	OCL detect
	Cause	OCL detect
	Measures	Please check the line (O/P Emergency Stop)
H0048	Error message	Faulty line (Mode Selector switch)
	Cause	The state of doubled line is not matching (Mode Selector switch)
	Measures	Please check the line (Mode Selector switch)
H0049	Error message	Faulty Line (T/B Enable Switch)
	Cause	The state of doubled line is not matching (T/B Enable Switch)
	Measures	Please check the line (T/B Enable Switch)
H0050	Error message	EMG signal is input. (external)
	Cause	The external emergency stop is being input. If the emergency stop of T/B turns on, this error may occur simultaneously.
	Measures	Release the external emergency stop signal.
H0051	Error message	Wiring of the external emergency stop is abnormal.
	Cause	If the emergency stop of T/B turns on, this error may occur simultaneously.
	Measures	Turn OFF the power supply. Confirm whether there is any problem in wiring of the external emergency stop switch. And, please confirm whether it is wiring of the dual line. Refer to the "Examples of safety measures" given in separate "Standard Specifications Manual" for external emergency stop switch wiring. Turn on the power supply again after checking.
H0053	Error message	EMG signal is input.(Add.Axis2)
	Cause	The external emergency stop to addition axis amplifier is inputting.
	Measures	Check the emergency stop of Additional Axis servo amp. Or the EM1 (forced outage) line of the addition axis may be open. Please confirm connection. In addition, the External Emergency Stop 1 and 2 are separated. The "External Emergency Stop 1" is for I/F card, and the "External Emergency Stop 2" is for main device of the amplifier
H0054	Error message	Faulty wiring (External EMG power)
	Cause	The state of doubled wiring is not matching (External EMG)
	Measures	Please check the wiring (External Emergency Stop)

Error No.	Error cause and measures	
H0061	Error message	EMG line is faulty.(O.Panel)
	Cause	The emergency stop line isn't stable.
	Measures	Confirm whether there is any problem in wiring of the external emergency stop switch. And, please confirm whether it is wiring of the double line. Refer to the "Examples of safety measures" given in separate "Standard Specifications Manual" for external emergency stop switch wiring. Turn on the power supply again after checking.
H0070	Error message	EMG signal is input. (T.Box)
	Cause	EMG signal is input. (T.Box) Or when using the UL specification, the brake release switch is turning ON.
	Measures	Cancel the T/B emergency stop. Check the emergency stop switch of teaching pendant.. Or when using the UL specification, turn OFF the brake release switch. If the alarm cannot be canceled, check the fuse of the safe unit (TZ348). If the fuse broke off, exchange new fuse.
H0071	Error message	EMG line is faulty.(T.Box)
	Cause	The emergency stop line isn't stable.
	Measures	Confirm whether there is any problem in wiring of the external emergency stop switch. And, please confirm whether it is wiring of the double line. Refer to the "Examples of safety measures" given in separate "Standard Specifications Manual" for external emergency stop switch wiring. Turn on the power supply again after checking.
H0074	Error message	Faulty line (T/B Enable/Disable).
	Cause	The state of doubled line is not matching (T/B Enable/Disable).
	Measures	Confirm whether T/B is connected correctly. If it comes back, contact to your service provider.
H0075	Error message	TB communication error
	Cause	Communication between the RC and TB was cut off.
	Measures	If it comes back, contact your service provider.
H0077 *	Error message	Electric double layer overvolt.
	Cause	A power supply of a substrate in the robot is broken
	Measures	Turn the power OFF and contact your service provider
H0083 *	Error message	Fuse is blown (hand)
	Cause	The pneumatic hand's power fuse has broken. Possibly the power supply line of the hand input/output signal short-circuited.
	Measures	Confirm that the hand input/output cables shown below are connected correctly with no short circuit caused by biting of the cables. ・Hand input cables (HC1 to HC8) and cables connected to them. ・Hand output cables (GR1 to GR8) and cables connected to them. Exchange the fuse. Refer to Page 65, "2.1.2 Place where hand fuse replacement is required" for details. If the error recurs after replacing the fuse, contact the manufacturer.
H0086	Error message	Hand module overcurrent
	Cause	The motorized hand's motor or circuit board has broken.
	Measures	Exchange the motorized hand's motor or circuit board.
H0090	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Fuse is blown (brake)
	Cause	A brake failure or a ground fault of brake cable may have caused the error.
	Measures	There are two brake fuses inside the robot controller. Refer to Page 64, "2.1.1 Place where a brake fuse replacement is required." and replace both the fuses. If it comes back after replacing the fuses, contact to your dealer.
	Error message	Fuse is blown (T/B)
	Cause	The cable may be disconnected or ground fault.
	Measures	Confirm whether there is any problem in cable. Turn on the power supply again after checking. If it comes back, contact to your dealer.
L0091	Error message	Can't access the Special signal
	Cause	The dedicated output signal is assigned to the specified signal. This signal cannot be used in duplicate.
	Measures	Confirm whether the same dedicated output number is assigned to the separate dedicated output signal. Change the output No., or change the dedicated output assignment parameter.

Error No.	Error cause and measures	
H0095	Error message	Brake release switch is turning on.
	Cause	Cannot execute while brake release switch is turning on.
	Measures	Please check the brake release switch and turn it off.
H0097 *	Error message	Abnormal MC status
	Cause	The duplex magnetic contactor status signal status is not consistent.
	Measures	The status of the duplex system for the magnetic contactor (MC) is not consistent. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer.
H0098 *	Error message	Abnormal SR status
	Cause	The duplex safety relay status signal status is not consistent.
	Measures	The status of the duplex system for the safety relay (SR) is not consistent. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer.
H0099 *	Error message	S/W ver. is illegal (**)
	Cause	Servo software with an old version is installed.
	Measures	Change to the correct version of software is necessary. Contact your service provider. * The correct version of servo software is shown in (**). * The error message and "software" of the cause are changed as follows according to the software in which an error occurs. CCPU main system, OS, RCPU sub system, CCPU sub system, servo S/W, FPGA OS, sub system, FPGA system, boot
C0099	Error message	R/C ver. is illegal (**)
	Cause	R/C with an old version is installed.
	Measures	Change to the correct version of software is necessary. Contact your service provider. * The correct version of software is shown in (**).
H0100 *	Error message	Temperature in RC is too high
	Cause	The intake fan is not operating, or the fan filter is clogged.
	Measures	Check the operation of the intake fan, or clean or replace the fan filter if necessary. Confirms that the environmental temperature is the specification range. When it comes back, contact to the dealer.
L0101	Error message	Temperature in RC is too high
	Cause	The intake fan is not operating, or the fan filter is clogged.
	Measures	Check the operation of the intake fan, or clean or replace the fan filter if necessary. Confirms that the environmental temperature is the specification range. When it comes back, contact to the dealer.
C0102	Error message	Temperature in RC is too high
	Cause	The intake fan is not operating, or the fan filter is clogged.
	Measures	Check the operation of the intake fan, or clean or replace the fan filter if necessary. Confirms that the environmental temperature is the specification range. When it comes back, contact to the dealer.
H0103 *	Error message	Temp. in robot CPU is too high
	Cause	Ambient temperature of robot CPU becomes high
	Measures	Please lower ambient temperature to the specification range
L0104	Error message	Temp. in robot CPU is too high
	Cause	Ambient temperature of robot CPU becomes high
	Measures	Please lower ambient temperature to the specification range
C0105	Error message	Temp. in robot CPU is too high
	Cause	Ambient temperature of robot CPU becomes high
	Measures	Please lower ambient temperature to the specification range
H0117 *	Error message	12V power supply error (brake)
	Cause	The output of the power supply deviated from the specified range
	Measures	Turn the power OFF and contact your service provider
C0120 *	Error message	Instantaneous power failure
	Cause	The power was OFF for 20msec or more
	Measures	Check the power supply connection and power supply state

Error No.	Error cause and measures	
H0130 *	Error message	The initialization connection error of system remote I/O.
	Cause	Communication line of System Remote I/O is illegal
	Measures	Confirm whether the communication cable is grounded correctly or connection correctly. In CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU when turning the power ON.
C0150	Error message	Undefined robot serial number
	Cause	Undefined robot serial number
	Measures	Input the robot serial number Refer to separate manual: "Controller setup, basic operation, and maintenance" for the setting method.
C0151	Error message	Undefined robot serial number
	Cause	Undefined robot serial number
	Measures	Input the robot serial number
C0152	Error message	Unmatched robot serial number
	Cause	Robot arm that connection was replaced
	Measures	Please confirm the connected robot body is not replaced. If replaced, correct the combination. If this error occurs after purchasing the robot arm or changing robot arm and controller combination, please input the serial number. Refer to "Input the serial number" in separate manual: "Controller setup, basic operation, and maintenance" for the setting method. After correcting the combination, please confirm that the programs and parameters you use are set to the robot controller before starting robot operation.
H016m (m=1 ～ 3)	Error message	Cannot use PIO I/F and Unit
	Cause	The I/O Channel number of PIO interface and PIO unit overlaps.
	Measures	Select either of Parallel I/O interface or Parallel I/O unit.
H017m (m=1 ～ 2)	Error message	Parallel I/O interface H/W error.
	Cause	The parallel-input/output card broke or the external power source for the parallel input/outputs was cut off.
	Measures	If it comes back, exchange the Parallel I/O interface card.
H018m (m=1 ～ 2)	Error message	Fuse is blown.(PIO) (Slot m-Fn)
	Cause	Fuse is blown.(Parallel I/O interface)
	Measures	Change fuse (Parallel I/O interface)
	Details	The electric fuse Fn (n= 1-4) of the parallel input output interface installed to the option slot m (m= 1-2) is open. Removes the cause by which the fuse open and replaces the fuse. 
H0210 *	Error message	Power supply error (***)
	Cause	The power supply output voltage is out of the specified range.
	Measures	The output voltage of the power supply in the robot controller is the specified value or higher/lower. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer. * The part where an error is detected is shown in "***" in the message. 24V, 5V, 3.3V, 2.5V, 1.8V, RCPU 1.5V, CCPU 1.5V, 1.2V, RCPU 1.15V, CCPU 1.15V, 1.1V, Ether 1V, ServoM 1V, ServoS 1V

Error No.	Error cause and measures	
H0211 *	Error message	Power supply error (***)
	Cause	The power supply output voltage is out of the specified range.
	Measures	The output voltage of the power supply in the robot CPU is the specified value or higher/lower. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer. * The part where an error is detected is shown in "***" in the message. 5V、3.3V、2.5V、1.8V、1.5V、1.3V、1.15V、1V、FPGA
H0212 *	Error message	Power supply error
	Cause	The output of the power supply deviated from the specified range.
	Measures	Turn the power OFF and ON once. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer.
H0213 *	Error message	Power supply error
	Cause	The output of the power supply deviated from the specified range.
	Measures	Turn the power OFF and ON once. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer.
H0220 *	Error message	Memory fault (main CPU)
	Cause	A memory fault is detected.
	Measures	A fault (inconsistent data between writing/reading, broken retention data) is detected for the memory (DRAM) to which the main CPU has access. After turning OFF the power, turn ON the power again to reset the error. If the same error recurs, contact the manufacturer.
H0230 *	Error message	Safety parameter error (xxxxx)
	Cause	Detect errors in safety parameter value
	Measures	The safety parameter setting is not correct. Check the parameter setting shown in the "(xxxxx)" part of the error message, and set an appropriate value.
H0231 *	Error message	Parameter CRC error (xxxxx)
	Cause	Detect CRC errors in safety parameter value
	Measures	Please check the safety parameter settings and write parameters.
C0240 *	Error message	Setting mismatch of safety function
	Cause	The safety function is disabled.
	Measures	Safety logic settings is set but safety function is disabled. Delete Safety Logic settings or Enable Safety Function.
H0241 *	Error message	Fault in Safety Communication
	Cause	Detect a fault in safety communication.
	Measures	Turn the power OFF and ON once If the same error recurs, contact the manufacturer.
H0242 *	Error message	Fault in Safety Data
	Cause	Detect a fault in safety data.
	Measures	Turn the power OFF and ON once If the same error recurs, contact the manufacturer.
H0243	Error message	XXX function is not available
	Cause	The Safety Function is not supported by this robot
	Measures	Disable the safety function
H0260 *	Error message	EMG circuit diagnosis error
	Cause	Voltage diagnosis error
	Measures	Turn the power OFF and ON once
H0270 *	Error message	Voltage diagnosis error
	Cause	Voltage diagnosis error
	Measures	Turn the power OFF and ON once
H0280 *	Error message	STO circuit diagnosis error
	Cause	STO citcuit error
	Measures	Turn the power OFF and ON once

Error No.	Error cause and measures	
H0290 *	Error message	PLD control error
	Cause	PLD control error
	Measures	Turn the power OFF and ON once
H0310	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Mecha board access error
	Cause	Failed in access to a Mecha board
	Measures	Please confirm the connection with Mecha board
	Error message	Mecha board buffer access error
	Cause	Cannot store more than 100 of the data to send/receive buffer
	Measures	Please confirm the communication with Mecha board
H0311 *	Error message	RemoteI/O unit config error
	Cause	The multiple RemoteI/O units which do not support are connected
	Measures	Please confirm the constitution of RemoteI/O unit When extended safety unit is connected, restart the unit.
H0315	Error message	Can't access the D-device area
	Cause	This area cannot be used in duplicate
	Measures	Please change the D-device start number in parameter DDEVVLn
H0316	Error message	The D-device area is out of range
	Cause	An area corresponding to the variable type can't be allocated
	Measures	Please change the D-device start number in parameter DDEVVLn
H0317	Error message	The D-device variable duplication
	Cause	Cannot be doubly allotted to one variable
	Measures	Please change the variable name in parameter DDEVVLn
C043n (n indicates the axis number (1 to 8).)	Error message	Servo amplifier motor overheat
	Cause	The motor or encoder's thermal protector activated.
	Measures	Reduce the speed and acceleration of the robot.
C049n * (n indicates the fan number (1 to 8).)	Error message	Alarm of fan in the robot
	Cause	Fan in the robot might be out of order
	Measures	Please exchange the fan in the robot
H050n * (n indicates the axis number (1 to 8).)	Error message	Servo axis setting error
	Cause	The setting of the axis number selection switch is illegal
	Measures	Confirm the setting of the axis selection switch
H0510 *	Error message	The converter setting is illegal
	Cause	The external emergency-stop input power was detected except external emergency-stop mode.
	Measures	Setting is wrong. When it comes back, contact to the dealer.
H0520 *	Error message	Robot axis setting illegal
	Cause	The setting of the servo axis used by the mechanism is duplicated with another mechanism's axis.
	Measures	Correctly set.
H053n * (n indicates the axis number (1 to 8).)	Error message	Servo sys. error (memory)
	Cause	The servo amplifier memory IC's check sum is illegal.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H054n * (n indicates the axis number (1 to 8).)	Error message	Servo sys. error (over run)
	Cause	The servo amplifier software data process did not end within the specified time.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H055n * (n indicates the axis number (1 to 8).)	Error message	Servo sys. error (mag. pole pos)
	Cause	An error was detected in the magnetic pole position detection signal of the detector.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.

Error No.	Error cause and measures	
H056n ＊ (n indicates the axis number (1 to 8).)	Error message	Servo sys. error (A/D)
	Cause	1) An error was found in the servo amplifier's A/D converter during initialization. 2) The stop signal or servo OFF signal was input when the door switch signal or EMG signal was input.
	Measures	1) Turn the power OFF and ON once. If it comes back, contact your service provider. 2) Do not input the door switch signal or EMG signal with another signal simultaneously.
H057n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder error (EEPROM)
	Cause	An error was detected in EEPROM data of the serial pulse encoder.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H058n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder error (LED)
	Cause	The LED of the serial pulse encoder has been deteriorated.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H059n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder error (position data)
	Cause	An error was detected in the position data within a single rotation of the encoder.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H060n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder no-signal detection 1
	Cause	An error was detected in the operating input of the detector mounted on the edge of the motor.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H061n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder no-signal detection 2
	Cause	An error was detected in the operating input of the detector mounted on the edge of the machine.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H062n ＊ (n indicates the axis number (1 to 8).)	Error message	Servo amplifier LSI error
	Cause	An operation error was detected in the LSI of the servo amplifier.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H063n ＊ (n indicates the axis number (1 to 8).)	Error message	Unused axis servo error
	Cause	A power module error occurred in an axis which not use the movement control.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H064n ＊ (n indicates the axis number (1 to 8).)	Error message	System error (ABS CPU)
	Cause	An error in the CPU of the absolute position linear scale was detected.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H065n ＊ (n indicates the axis number (1 to 8).)	Error message	Absolute position error
	Cause	An error was detected in the absolute position detection circuit within the absolute position linear scale.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H066n ＊ (n indicates the axis number (1 to 8).)	Error message	Incremental position error
	Cause	An error was detected in the relative position detection circuit within the absolute position linear scale.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.

Error No.	Error cause and measures	
H067n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder CPU error
	Cause	An error was detected in the CPU of the position detector.
	Measures	Turn the power OFF and ON once. Also, carefully check whether there is no deviation in the operating position of the robot. If it is deviated, set the origin position (OP) again. For more information about the operating procedure, refer to the separate volume, "Instruction Manual/Robot Arm Setup to Maintenance." If it comes back, contact your service provider.
H068n ＊ (n indicates the axis number (1 to 8).)	Error message	Encoder LED error
	Cause	Deterioration of the position detector's LED was detected.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0690 ＊	Error message	Regeneration circuit error
	Cause	A regenerative transistor or resistor error was detected. This error may also occur when the power supply voltage is high.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0700 ＊	Error message	P.S. external contactor fusing
	Cause	The contactor was turned ON even though READY is OFF.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0710 ＊	Error message	Servo amp. relay error.
	Cause	The relay on servo cpu card did not turn ON.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0711 ＊	Error message	Discharge resistance relay error
	Cause	Discharge resistance circuit on converter is state-of-discharge
	Measures	Turn the power OFF and ON once
H0712 ＊	Error message	Converter fuse fusing
	Cause	Fuse of a converter fused for a ground fault or short circuit
	Measures	Please exchange a fuse of a converter Investigate and correct the ground fault or short circuit portion in the wiring made by the customer. Then, replace the 3.2 A fuse inside the controller. Refer to Page 67, "2.1.4 Place where a converter fuse replacement is required", for details. (On details of the 3.2 A fuse, contact the manufacturer.) If no improvement is made after carrying out the above measures, please contact the manufacturer.
H0713 ＊	Error message	Encoder power supply fuse trip
	Cause	Encoder power supply fuse tripped by ground fault, short circuit
	Measures	Turn the power OFF, wait a while, and then turn ON again
H0720 ＊	Error message	Power supply watch dog
	Cause	The converter software process did not end within the specified time.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0730 ＊	Error message	Power supply rush relay fusing
	Cause	The rush resistance short-circuit relay did not turn OFF.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0740 ＊	Error message	Power supply main circuit error
	Cause	The charge operation of the main circuit capacitor is not normal. Connection of the external emergency stop has the mistake.
	Measures	Turn the power OFF and ON once. Confirm that the power supply voltage is in the specification value and the connection of the external emergency stop is correct. If it comes back, confirm the time of occurrence of this error being "servo ON/OFF", or being "power supply OFF/ON", and contact your service provider
H0742	Error message	Power supply main circuit error
	Cause	A main circuit voltage has decreased because of a failure of the Safety relay on a converter card.
	Measures	Turn the power OFF and ON once. Confirm whether there is any problem in wiring of the external emergency stop switch. Refer to the "Examples of safety measures" given in separate "Standard Specifications Manual" for external emergency stop switch wiring. If it comes back, contact to your service provider.

Error No.	Error cause and measures	
H0743 *	Error message	Power supply main circuit error3.
	Cause	A main circuit voltage has decreased because of contactor fail.
	Measures	Turns off the power once and turns on on the power supply again. Confirm power supply voltage and the connection of the external emergency stop. When it comes back, contact to the dealer.
H0750 *	Error message	Power supply memory error
	Cause	An error in the memory circuit of converter or AD converter was detected.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0760 *	Error message	Power supply error
	Cause	An error was detected in the data comm. with the power supply
	Measures	Turn the power OFF and ON once
H0770 *	Error message	Power supply process error
	Cause	An error occurred in the process cycle of power supply
	Measures	Turn the power OFF and ON once. Check that there is not a source of noise. If it comes back, contact your service provider.
H078n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier watch dog
	Cause	The servo amplifier software process is not operating correctly.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H079n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier board error
	Cause	An error was detected in the servo amplifier's PCB.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H080n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier clock error
	Cause	An error was detected in the servo amplifier's clock.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H081n * (n indicates the axis number (1 to 8).)	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Servo amplifier under voltage
	Cause	The PN bus voltage dropped to 200V or less. Momentary power failure may have occurred.
	Measures	Check the primary voltage.
	Error message	Voltage error at accel/decel
	Cause	A motor control error was detected due to an input voltage drop
	Measures	Turn the power OFF and ON once. Check the primary voltage.
H0820 * H082n * (n indicates the axis number (1 to 8).)	Error message	Motor ground fault
	Cause	A motor ground fault was detected. A connection or conductance error may have occurred in the motor cable.
	Measures	Turn the power OFF and ON once. Check connection and continuity of motor cable.
H083n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier overvoltage
	Cause	The PN bus voltage rose to 400V or more.
	Measures	Check the primary power supply voltage. Turn the power OFF and ON once.
H0840 *	Error message	Instantaneous power failure (SRV)
	Cause	A power shutdown status of 50 msec or longer has occurred.
	Measures	Check the power voltage. Turn the power OFF and ON once.
H0850 *	Error message	Power supply voltage incorrect
	Cause	The input power (L1, L2, L3) has an open phase, the voltage is not within the specifications, or the 100V/200V specifications changeover setting is incorrect.
	Measures	Check the power connection, power state or the setting.
H0860	Error message	Power supply overvoltage
	Cause	The voltage across the converter's L+ and L− exceeded 410V.
	Measures	Check the power supply connection and power supply state.
H0870 *	Error message	Power module overheat (Fan Stop)
	Cause	A cooling fan stopped, and overheat occurred
	Measures	Check the rotation of fan, and the connector connected to a fan

Error No.	Error cause and measures	
H0880 * H088n * (n indicates the axis number (1 to 8).)	Error message	Power module overheat
	Cause	Overheating of the power module regenerative resistor was detected.
	Measures	Turn the controller power OFF, wait a while, and then turn ON again. If it comes back, contact your service provider.
H089n (n indicates the axis number (1 to 8).)	Error message	Servo amplifier motor overheat
	Cause	The position detector's thermal protector activated.
	Measures	Turn the controller power OFF, wait a while, and then turn ON again. Decrease the acceleration/deceleration time of the operation speed, for instance. Refer to "Detailed explanation of command words"/"Accel (Accelerate)," "Ovrd (Override)" and "Spd (Speed)," or "Detailed explanation of Robot Status Variable"/"M_SetAdl," "M_LdfAct" and "Functions set with parameters"/"JADL (Optimum acceleration/deceleration adjustment rate)" of the Separate Volume, "INSTRUCTION MANUAL/Detailed Explanation of Functions and Operations."
H090n * (n indicates the axis number (1 to 8).)	Error message	Absolute position overspeed
	Cause	It moved 45 mm/sec or faster with the absolute position linear scale during initialization.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H091n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier overspeed
	Cause	A speed exceeding the motor's tolerable speed was detected.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H0920 * H092n * (n indicates the axis number (1 to 8).)	Error message	Power module overcurrent
	Cause	A servo amplifier or power supply overcurrent was detected. An error was detected in the servo amplifier's gate circuit. A connection or conductance error may have occurred in the motor cable.
	Measures	Confirms the connection of the machine cable and the locomotion-axis cable. If it comes back, contact your service provider.
H093n * (n indicates the axis number (1 to 8).)	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Motor overcurrent
	Cause	An excessive current flowed to the motor, or the A/D converter output is abnormal. An abnormality may have occurred in the connection of the motor's power line.
	Measures	Turn the power OFF and ON once. Confirms the connection of the machine cable and the locomotion-axis cable etc. If it comes back, contact your service provider.
	Error message	Motor overcurrent (Grounding)
	Cause	The motor power cable is in contact with ground
	Measures	Turn the power OFF and ON once. Confirms the connection of the machine cable and the locomotion-axis cable etc. If it comes back, contact your service provider.
H094n (n indicates the axis number (1 to 8).)	Error message	Overload (over weight 1)
	Cause	Operation tight for a motor (operation with high duty) was performed more than fixed time.
	Measures	Decrease the acceleration/deceleration time of the operation speed, for instance. Refer to "Detailed explanation of command words"/"Accel (Accelerate)," "Ovrd (Override)" and "Spd (Speed)," or "Detailed explanation of Robot Status Variable"/"M_SetAdl," "M_LdfAct" and "Functions set with parameters"/"JADL (Optimum acceleration/deceleration adjustment rate)" of the Separate Volume, "INSTRUCTION MANUAL/Detailed Explanation of Functions and Operations." Confirms that conveyance conditions (hand mass, work-piece mass) are less than specification values. When it comes back, contact to the dealer.
H095n (n indicates the axis number (1 to 8).)	Error message	Overload (over weight 2)
	Cause	The maximum output current continued for more than one second.
	Measures	Check the load weight and the robot pressing, etc. Confirms that conveyance conditions (hand mass, work-piece mass) are less than specification values. When it comes back, contact to the dealer.

Error No.	Error cause and measures	
H096n (n indicates the axis number (1 to 8).)	Error message	Excessive error 1
	Cause	The position error exceeded at servo ON. Moreover, this error may occur during the emergency-stop deceleration.
	Measures	・Check the load weight and press, etc. Confirms the connection of the machine cable and the locomotion-axis cable etc. If the surrounding temperature is low, or starting after stopping operation for an extended period of time, perform running-in operation at low speed or use the warm-up operation mode. ・ When hand offset is long and acceleration-and-deceleration control is fixed or tracking is active in RH-20FRH series. please reduce the acceleration and deceleration speeds (Accel command) and movement speed (Ovrd command). Refer to separate "Instruction Manual/Detailed Explanation of Functions and Operations" for details of each command. (This error may occur during the emergency-stop deceleration.) ・While operation is performed in the compliance mode of the joint coordinate system, if the Excessive error 1 (H096n) occurs, increase the set value of parameter CMPJCLL to suppress this error. Refer to "Detailed explanation of command words"/"Cmp Jnt (Compliance Joint)" and "Movement parameter"/"CMPJCLL (Current Limit Level for Cmp Jnt)" given in separate "INSTRUCTION MANUAL/ Detailed explanation of functions and operations".
H097n (n indicates the axis number (1 to 8).)	Error message	Excessive error 2
	Cause	The position error exceeded at servo OFF.
	Measures	Check the moving robot arm by something power. When it comes back, contact to the dealer.
H098n (n indicates the axis number (1 to 8).)	Error message	Excessive error 3
	Cause	Abnormal motor power line connection.
	Measures	Check the connection of motor power line. When the excessive error 1 was detected, the current of the motor is off.
H101n (n indicates the axis number (1 to 8).)	Error message	Collision detection
	Cause	A collision was detected.
	Measures	1) If the robot has stopped by interference with peripheral equipment, move the arm to part from peripheral equipment using jog operation. Depending on the level of collision, the collision detection error may occur again. In that case, turn on the servo power again and do jog operation. If it still recurs, release the brake and move the arm by hand. 2) If this error occurs without having collided, please adjust the collision detection level. If collision is detected incorrectly during automatic operation, enlarge the setting value of the parameter (COLLVL) corresponding to axis. If collision is detected incorrectly during jog operation, enlarge the setting value of the parameter (COLLVLJG) corresponding to axis. However, since the detection level drops by enlarging the set value, don't enlarge too much. And, the incorrect detection can be reduced when setup value of parameter (HNDDATn、WRKDATn) is correct. 3) If the speed excessive error has occurred at the same time, the torque alteration by rapid speed change may be detected as a collision state. Remove other causes of the error and confirm movement again. 4) In case of operation under the environment of low temperature or after the long term stoppage, the collision detection error may occur by viscous transmutation of the grease used. In such a case, operate by accustoming at low speed (warm-up), or use the warm-up operation mode.
H102n (n indicates the axis number (1 to 8).)	Error message	Servo AMP over-regeneration
	Cause	The additional axis exceeded the regenerative performance limit.
	Measures	Check the regenerative capacity and parameters for the additional axis. If it comes back, contact your service provider. Regeneration resistance may be disconnected.
H1030 *	Error message	Power supply over-regeneration
	Cause	The converter's regenerative performance limit was exceeded.
	Measures	Wait at least 15 minutes in the power ON state, and then turn the power OFF and ON. If it comes back, contact your service provider. Regeneration resistance may be disconnected.
H104n * (n indicates the axis number (1 to 8).)	Error message	Encoder init communication error
	Cause	An abnormality may have occurred in the position detector cable connection.
	Measures	1) Turn the power OFF and ON once. If it comes back, contact your service provider. 2) When instantaneous power failure errors occur simultaneously, turn on the power supply again.
H105n * (n indicates the axis number (1 to 8).)	Error message	Encoder init communication error
	Cause	Initial communication with the encoder was not possible
	Measures	Check the encoder cable connection

Error No.	Error cause and measures	
H106n * (n indicates the axis number (1 to 8).)	Error message	Encoder communication error
	Cause	Communication between the encoder and detector was cut off
	Measures	Check the encoder cable connection
H107n * (n indicates the axis number (1 to 8).)	Error message	Encoder communication error
	Cause	Communication with the position detector was cut off.
	Measures	Confirms the signal cable of the machine cable and the locomotion-axis cable etc. If it comes back, contact your service provider.
H108n * (n indicates the axis number (1 to 8).)	Error message	Servo AMP communication error
	Cause	An abnormality may have occurred in the communication cable connection.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider. CR800-R/CR800-Q: Check the connection between the robot CPU system and the controller.
H1090 * H109n * (n indicates the axis number (1 to 8).)	Error message	Servo AMP initialization error
	Cause	An abnormality may have occurred in the servo axis settings (parameters, rotary switches).
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider. CR800-R/CR800-Q: Check the connection between the robot CPU system and the robot controller. And when turning the power ON, turn on the switch of the robot controller, then of the robot CPU. Using the additional axis: Confirms the axis setting switch, cable connection with robot controller, setting parameter, condition of power supply of additional axis amplifier, type of additional axis amplifier etc. When applying a power supply, turn on the additional axis amplifier first, then turn on the controller. And when turning the power ON, turn on the switch of the drive unit, then of the robot CPU.
H1100 *	Error message	Servo com. receive error
	Cause	Abnormality occurred in data reception from a servo amplifier
	Measures	Check the communication cable connection and conductivity
H1110 *	Error message	Servo communication timeout
	Cause	Data from a servo amplifier is not received
	Measures	Check the communication cable connection and conductivity
H111n (n indicates the axis number (1 to 8).)	Error message	SRV-AMP comm. error
	Cause	A communication error was detected between the servo amplifier and PC.
	Measures	Check the communication cable connection and conductivity. If it comes back, contact your service provider.
H112n * (n indicates the axis number (1 to 8).)	Error message	Encoder ABS position data lost
	Cause	The absolute position data in the position detector was lost. The voltage of the robot-arm or additional axis's backup battery may be dropping.
	Measures	Please set up the origin by ABS method after replacing the batteries. (Since position data can be perfectly restored if the origin is set up by the ABS method, re-teaching is unnecessary.) Refer to separate "Instruction Manual/ROBOT ARM SETUP & MAINTENANCE" for ABS method. If the voltage of robot battery is low, reset the error, and set up the origin by ABS method. Then, the robot can be operated without replacing the battery. However, this error occurs again when the controller is re-turned on. It is recommended to replace the battery at the earliest opportunity.
H113n * (n indicates the axis number (1 to 8).)	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Encoder per rotation data error
	Cause	An error was detected in the position detector's one rotation data.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
	Error message	Encoder data scattered
	Cause	Position deviation by scattered encoder data occurred
	Measures	Turns off the power supply once and turns on again. Check that there is not a source of excessive vibration. If it comes back, contact your service provider.
H114n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. data error (CRC)
	Cause	A CRC error was detected in the data from servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.

Error No.	Error cause and measures	
H115n * (n indicates the axis number (1 to 8).)	Error message	Large command position
	Cause	The command position from the RC is abnormally large.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H116n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. error (frame)
	Cause	An error was detected in the communication frame from servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H117n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. error (info)
	Cause	An error was detected in the communication information from the RC.
	Measures	Check the communication cable connection and conductivity. If it comes back, contact your service provider.
H118n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier feedback error 1
	Cause	Pulses skipped in the position detector's feedback signal.
	Measures	Check the detector cable connection and conductivity. If it comes back, contact your service provider.
H119n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier feedback error 2
	Cause	Displacement occurred in the feedback amounts between the detectors on the motor edge and on the machine edge.
	Measures	Check the detector cable connection and conductivity. If it comes back, contact your service provider.
H1200 *	Error message	SRV-AMP Comm. data error (CRC)
	Cause	A CRC error was detected in the communication data from the servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H121n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. data error (ID)
	Cause	A data ID error was detected in the communication data from the servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H122n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. data error (axis No)
	Cause	An axis No. error was detected in the communication data from the servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H123n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP Comm. data error (SubID)
	Cause	A Sub ID error was detected in the communication data from the servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H1240 *	Error message	SRV-AMP Comm. data error (frame)
	Cause	An No. of received frames error was detected in the communication data from the servo amplifier.
	Measures	Turns off the power supply once and turns on again. If it comes back, contact your service provider.
H125n * (n indicates the axis number (1 to 8).)	Error message	Servo amplifier parameter error
	Cause	An error was detected in the servo parameter.
	Measures	Confirms whether the type displayed on the rated name plate of controller and T/B is the same. If different, restores the type data (backup data). If it comes back, contact your service provider.
C126n (n indicates the axis number (1 to 8).)	Error message	Encoder communication error
	Cause	Initial communication could not be established with the low-speed serial type absolute position linear scale.
	Measures	Check the detector cable connection and conductivity. If it comes back, contact your service provider.
C127n (n indicates the axis number (1 to 8).)	Error message	Encoder communication error
	Cause	The serial data of absolute position was abnormally transmitted.
	Measures	Check the detector cable connection and conductivity. Moreover, confirms carefully that the moving position of the robot has not deviated, and if it has deviated, sets up the origin again. Refer to separate "Instruction Manual/ROBOT ARM SETUP & MAINTENANCE" for operation method. If it comes back, contact your service provider.
C128n (n indicates the axis number (1 to 8).)	Error message	Encoder serial format error
	Cause	Absolute position serial data format was incorrect.
	Measures	Check the detector cable connection and conductivity. Moreover, confirms carefully that the moving position of the robot has not deviated, and if it has deviated, sets up the origin again. Refer to separate "Instruction Manual/ROBOT ARM SETUP & MAINTENANCE" for operation method. If it comes back, contact your service provider.

Error No.	Error cause and measures	
C129n (n indicates the axis number (1 to 8).)	Error message	Absolute position fluctuation
	Cause	The absolute position data fluctuated when the power was turned ON.
	Measures	Check whether the axis moved due to arm dropping or external force when the power was turned ON.
C130n (n indicates the axis number (1 to 8).)	Error message	Servo AMP MP scale F/B error
	Cause	Excessive displacement was detected in the feedback amounts between the detector and the MP scale.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
C131n (n indicates the axis number (1 to 8).)	Error message	Servo AMP MP scale offset error
	Cause	Excessive displacement was detected in the feedback amounts between the detector and the MP scale.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
C132n (n indicates the axis number (1 to 8).)	Error message	Multi-rotation data error
	Cause	An error was detected in the position detector's multi-rotation data.
	Measures	When the power supply is turned on next time, the current position data may not be correctly detected. Please re-turn on the controller power supply and set up the origin by ABS method. Refer to separate "Instruction Manual/ROBOT ARM SETUP & MAINTENANCE" for ABS method. (Even if this warning occurs, unless the controller is re-turned on, it can operate perfectly. However, when re-turning on the controller, the position gap may occur. Please be sure to set up the origin by ABS method after re-turning on the controller power supply. Current position data returns to the normal position before error occurrence.) When it comes back, contact to the dealer.
C133n (n indicates the axis number (1 to 8).)	Error message	Encoder battery voltage low
	Cause	The battery voltage supplied to the position detector dropped.
	Measures	Replace the backup battery. For more information about the replacement procedure, refer to the separate volumes, "Instruction Manual/Robot Arm Setup to Maintenance". Even if this warning occurs, unless the controller is re-turned on, it can operate perfectly. However, if the battery consumption is intense, the Encoder ABS position data lost error (H112n) may occur when the controller is re-turned on. It is recommended to replace the battery at the earliest opportunity.
C134n (n indicates the axis number (1 to 8).)	Error message	Over-regeneration warning
	Cause	The regenerative level of the additional axis has risen to 80% or more.
	Measures	Check the regenerative capacity and parameters for the additional axis. If it comes back, contact your service provider. Regeneration resistance may be disconnected.
C135n (n indicates the axis number (1 to 8).)	Error message	Overload warning
	Cause	The overload level reached 95% or more.
	Measures	Check the load weight and the robot for collisions, etc.
H136n * (n indicates the axis number (1 to 8).)	Error message	Absolute position counter error
	Cause	The counter of absolute position is illegal.
	Measures	Confirm whether the connection of encoder cable and the battery voltage of arm is falling.
C137n (n indicates the axis number (1 to 8).)	Error message	Illegal parameter (servo)
	Cause	A parameter was set exceeding the setting range.
	Measures	The parameter has not been changed. Reset the correct value. If it comes back, contact your service provider.
C138n (n indicates the axis number (1 to 8).)	Error message	Removing control axis (servo)
	Cause	An instruction to remove the axis was issued by the controller.
	Measures	Cancel the instruction of removing axis.
H1390	Error message	Emergency stop (Servo amplifier)
	Cause	The emergency stop command has been input from the RC
	Measures	Release the emergency stop state
C1400	Error message	regeneration(AUX) frequency over
	Cause	Regeneration at the limit has occurred frequently
	Measures	Check the regeneration capacity
H1410 *	Error message	Instantaneous power interruption
	Cause	The power was momentarily interrupted
	Measures	Check power specification, and wiring of power and external emergency stop

Error No.	Error cause and measures	
C1420	Error message	Over-regeneration warning
	Cause	The regeneration level reached 80% or more.
	Measures	Lower the robot's movement speed. If it comes back, contact your service provider. Regeneration resistance may be disconnected.
C1430	Error message	Servo amplifier main circuit OFF
	Cause	The servo turned ON while the main circuit power was OFF
	Measures	Turn the main circuit power ON
H144n * (n indicates the axis number (1 to 8).)	Error message	System error (servo 2)
	Cause	Trouble occurred in the current processing processor.
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H1450 *	Error message	Instantaneous power interruption
	Cause	The power was momentarily interrupted at the power supply
	Measures	Check power specification, and wiring of power and external emergency stop
H1460 *	Error message	Power supply overcurrent
	Cause	Overcurrent in the power module in the power supply.
	Measures	Check the AC power line. If it comes back, contact your service provider.
H1470 *	Error message	Frequency error
	Cause	AC frequency is out of range.
	Measures	Check the AC power line frequency. If it comes back, contact your service provider.
H148n * (n indicates the axis number (1 to 8).)	Error message	Power supply parameter error
	Cause	Illegal at the power supply parameter
	Measures	Turn the power OFF and ON once. If it comes back, contact your service provider.
H1490 *	Error message	Power supply　Power module overheat.
	Cause	The temperature protection function of the power module operated.
	Measures	Clean up or replaces the air filters. When it comes back, contact to the dealer.
H1491	Error message	Illegal converter thermal servo AMP.
	Cause	Overheating of servo AMP converter.
	Measures	Turn the power OFF, wait a while, and then turn ON again.
H1492 *	Error message	Power supply converter overheat
	Cause	Overheating of the converter was detected
	Measures	Turn the power OFF, wait a while, and then turn ON again
H1493 *	Error message	Rush resistance overheat
	Cause	Overheating of the rush resistance was detected
	Measures	Turn the power OFF, wait a while, and then turn ON again
H1494 *	Error message	Discharge resistance overheat
	Cause	Overheating of the discharge resistance was detected
	Measures	Turn the power OFF, wait a while, and then turn ON again
H150n * (n indicates the axis number (1 to 8).)	Error message	Motor combination error
	Cause	The servo motor which does not correspond to addition axis servo amplifier is connected.
	Measures	Confirm the specifications of servo amplifier and the motor.
H151n * (n indicates the axis number (1 to 8).)	Error message	SRV-AMP USB comm. error
	Cause	A communication error was detected between the servo amp and PC
	Measures	Check the communication cable connection and conductivity
H152n * (n indicates the axis number (1 to 8).)	Error message	Servo motor output watt over
	Cause	The output wattage of the servo motor exceeded ratings
	Measures	Lower the rotation speed of the servo motor
H154n * (n indicates the axis number (1 to 8).)	Error message	Communication error between units
	Cause	The transmission data between servo amplifier units is illegal
	Measures	Check the communication cable connection and conductivity

Error No.	Error cause and measures	
H1550 *	Error message	Emergency stop is a breakdown
	Cause	The external emergency stop input is illegal
	Measures	Turn the power OFF and ON once. When it comes back, contact to the dealer.
H156n (n indicates the axis number (1 to 8).)	Error message	Excessive error 4
	Cause	The axis moved while executing servo ON processing.
	Measures	If it comes back, contact your service provider.
H157n (n indicates the axis number (1 to 8).)	Error message	Non-registered servo error. (This error may be caused by the error on the amplifier for additional axis.)
	Cause	A non-registered servo alarm occurred.
	Measures	Confirms the code by LED of servo additional axis amplifier, and refer to the instruction manual of servo amplifier. If the alarm cannot be reset, turn the power OFF and ON. If it comes back, contact your service provider.
C158n (n indicates the axis number (1 to 8).)	Error message	Non-registered servo warning. (This caution may be caused by the warning on the amplifier for additional axis.)
	Cause	A non-registered servo warning occurred.
	Measures	Confirms the code by LED of servo additional axis amplifier, and refer to the instruction manual of servo amplifier. If the alarm cannot be reset, turn the power OFF and ON. If it comes back, contact your service provider.
H1600 *	Error message	Mechanism un-setting.
	Cause	The mechanism is not set up.
	Measures	Set up one or more of mechanism. If this alarm occur after the restoring the data of controller check the restored data. If it comes back, contact your service provider.
H1601 *	Error message	Unmatched robot model
	Cause	Robot arm that connection is not correct
	Measures	Please confirm the connected robot arm
H1610 *	Error message	System error (illegal MEMECH)
	Cause	The mechanism module name is illegal or not registered.
	Measures	Correctly set. If it comes back, contact your service provider. If it comes back, contact your service provider.
C1620	Error message	Illegal robot No.
	Cause	When specifying the mechanism number by the external communications protocol, the mechanism number not existing was specified.
	Measures	Specify the existing mechanism number.
C1630	Error message	Cannot servo ON (during error)
	Cause	The servo cannot be turned ON during a servo error.
	Measures	Reset the servo error before turning the servo ON.
C1640	Error message	Cannot servo ON (DEADMAN OFF)
	Cause	The servo cannot be turned ON while the enable switch is OFF.
	Measures	Turn the enable switch ON before turning the servo ON.
C1650	Error message	Cannot servo ON (brake OFF)
	Cause	The servo cannot be turned ON when there is an axis with the brakes released.
	Measures	Lock the brakes for all axes before turning the servo ON.
C1660	Error message	Cannot servo ON (SRVON process)
	Cause	The servo cannot be turned ON during the servo ON process.
	Measures	Operate it after servo ON process is finished.
C1670	Error message	Cannot servo ON (SRVOFF process)
	Cause	The servo OFF process is being carried out.
	Measures	Operate it after servo OFF process is finished.

Error No.	Error cause and measures	
H1680	Error message	Cannot servo ON (timeout)
	Cause	The servo did not turn ON within the specified time.
	Measures	Turn the power supply OFF and ON once. If it comes back, confirm that power supply voltage is in the specification value, and connection of the external emergency stop is correct. And, when using the addition axis, confirm that the power supply voltage to the servo amplifier of addition axes is in the specification value and connection of AXMC is correct. If it comes back, contact your service provider.
H1681	Error message	Unexpected servo OFF
	Cause	The servo turned OFF unexpectedly.
	Measures	Confirm that primary power supply voltage is in the specification value and wiring of the external emergency stop line is correct. When using the additional axis, confirms whether the alarm occur on the additional axis. If it comes back, contact your service provider.
H1682	Error message	Servo ON Timeout (Safety relay).
	Cause	A main circuit voltage did not rise because of a failure of the Safety relay on a converter card.
	Measures	Turn the power OFF and ON once. Confirm that wiring of the external emergency stop line is correct. Confirm whether there is any problem in wiring of the external emergency stop switch. Refer to the ″Examples of safety measures″ given in separate ″Standard Specifications Manual″ for external emergency stop switch wiring. Please confirm whether in use of addition axis, there is any failure in the servo amplifier for addition axes. If it comes back, contact to your service provider.
H1683	Error message	Servo ON Timeout. (Contactor)
	Cause	A main circuit voltage did not rise because of contactor welded.
	Measures	Turns off the power supply once and turns on again. Confirm that primary power supply voltage is in the specification value and wiring of the external emergency stop line is correct. When it comes back, contact to the dealer.
C1690	Error message	Cannot brake operation (DEADMAN)
	Cause	The servo cannot be turned ON while the enable switch is OFF.
	Measures	Turn the enable switch ON before turning the servo ON.
C1700	Error message	Cannot brake operation (EMG)
	Cause	The brakes cannot be released while the emergency stop is input.
	Measures	Release the emergency stop state before operating.
C1710	Error message	Cannot brake operation (SRVON)
	Cause	The brakes cannot be operated during servo ON.
	Measures	Turn the servo OFF before operating.
C1720	Error message	Cannot brake operation (BRK OFF)
	Cause	The brakes cannot be released during the brake release process.
	Measures	Operate it after brake is released.
C1730	Error message	Cannot brake operation (BRK ON)
	Cause	The brakes cannot be locked during the brake lock process.
	Measures	Operate it after brake is locked.
C1740	Error message	Servo parameter change failure
	Cause	Other parameters cannot be changed during the parameter change process.
	Measures	Carry out the parameter change process again.
C1750	Error message	Servo parameter change failure
	Cause	Changes of the servo parameter failed.
	Measures	Carry out the parameter change process again.
C1760	Error message	Illegal origin data
	Cause	The origin setting data is not correct.
	Measures	Set the correct origin setting data. Confirm mistakes such as ″1(one)″ and ″I(alphabet)″, or ″O(alphabet)″ and ″0(zero)″, etc.

Error No.	Error cause and measures	
C1761	Error message	Illegal origin data in robot arm
	Cause	The origin data is illegal in robot arm
	Measures	Setting the origin
C1770	Error message	Origin setting incomplete
	Cause	The origin is not set.
	Measures	Re-execute after setting the origin.
C1780	Error message	Cannot set origin (illegal axis)
	Cause	The origin was not set simultaneously for the interference axis.
	Measures	Set the origin simultaneously for the interference axis. For example, the J3 axis and the J4 axis of the RH type robot. Please refer to the section of origin setting of separate manual: "ROBOT ARM SETUP & MAINTENANCE" for details.
C1781	Error message	Cannot set origin (SRVON)
	Cause	The origin was set during servo ON.
	Measures	Turn the servo OFF before setting the origin.
H179n *n (n indicates the axis number (1 to 8).)	Error message	Illegal parameter (MEJAR)
	Cause	The parameter (MEAJAR) setting is illegal. Or the setting value of the parameter related to addition axis control have exceeded the controllable operating range.
	Measures	The useful range of the operating range setting parameter (MEAJAR) is -131072.00 to +131072.00. Correct, if the set value is over the range. Or confirm the setting value of the parameter related to addition axis control.
H1800 *	Error message	Illegal parameter (MEMAR)
	Cause	The ABS operation range setting parameter MEMAR setting is incorrect. (Minus side value is larger than "0", or plus side value is smaller than "0")
	Measures	Set the value of the parameter "MEMAR" within the limits.
H1810 *	Error message	Illegal parameter (USERORG)
	Cause	The user origin setting parameter USERORG setting is incorrect.
	Measures	Correct the parameter USERORG.
L182n (n indicates the axis number (1 to 8).)	Error message	Pos. data disagree.Check origin
	Cause	Position data changed during power off.
	Measures	Check the origin, re-install if shifting. Re-set up the origin only for the axis which deviated by the ABS method. Refer to separate "Instruction Manual/ROBOT ARM SETUP & MAINTENANCE" for ABS method.
L1830	Error message	JRC. Exseeds the Pos. limit
	Cause	The JRC instruction exceeding the operation range was executed.
	Measures	Check the current position and the operating range.
L184n (n indicates the axis number (1 to 8).)	Error message	JRC Qtt. Setting Too Large
	Cause	The movement amount setting of the JRC is excessive.
	Measures	Correct the JRCQTT parameter.
C1850	Error message	Instantaneous power failure
	Cause	There was the momentary power failure.
	Measures	Check the power supply connection and power supply state.
L1860	Error message	Illegal parameter (TLC)
	Cause	The setting of the TLC parameter that sets the approach direction was incorrect.
	Measures	Correct the TLC parameter. (=X/Y/Z)
L1864	Error message	FTP parameter setting error (**) Note) "**" is substituted with the "parameter name".
	Cause	The FTP communication parameter setting lies outside the range.
	Measures	Check the setting and correct.
H188n * (n indicates the axis number (1 to 8).)	Error message	Jn addition axis amplifier cooling fan stop.
	Cause	The cooling fan of the addition axis amplifier of Jn axis may be out of order.
	Measures	Please replace the cooling fan of addition axis amplifier. * Please also refer to the instruction manual of the servo amplifier of usage.

Error No.	Error cause and measures	
C189n ＊ (n indicates the axis number (1 to 8).)	Error message	Jn addition axis amplifier cooling fan rev fall.
	Cause	The cooling fan of the addition axis amplifier of Jn axis may be out of order.
	Measures	Please replace the cooling fan of addition axis amplifier. ＊Please also refer to the instruction manual of the servo amplifier of usage.
C1940	Error message	Agitating fan stopped
	Cause	The fan for agitating inside the robot controller has stopped.
	Measures	Confirm rotation of the target cooling fan, and if out of order, replace them. Please refer to Page 68, "2.2 Fan installation place of robot controller." for the fan's mounting place.
H195n	Error message	Additional axis AMP error xx
	Cause	Alarm of the addition axis servo amplifier was detected.
	Measures	Refer to the instruction manuals of addition axis servo amplifier for the details of alarm. "xx" of the error message corresponds to the alarm number of addition axis servo amplifier (MR-J4- ☐ B). (The error reset methods differ for each alarm number of addition axis servo amplifier.)
C196n	Error message	Additional axis AMP Warning xx
	Cause	Warning of the addition axis servo amplifier was detected.
	Measures	Refer to the instruction manuals of addition axis servo amplifier for the details of warning. "xx" of the error message corresponds to the warning number of addition axis servo amplifier (MR-J4- ☐ B).
C1970	Error message	Agitating fan stopped (robot)
	Cause	The agitating fan in the robot stopped
	Measures	Confirm the rotation of the agitating fan ＊According to the type of the stopped fan, "inside the robot arm" and "inside the robot base" are shown separately.
L2000	Error message	The servo is OFF
	Cause	Because servo is turned off, the robot can't move.
	Measures	Turn the servo ON and then restart.
L2010	Error message	Pulse output was not possible
	Cause	There is an error in the pulse output designation.
	Measures	Correct the program.
L2020	Error message	Reading external position data
	Cause	A command that cannot be executed while reading the external commands was executed.
	Measures	Correct the program.
L2030	Error message	JOG operation cannot be accepted
	Cause	The JOG operation request was issued when the JOG operation request could not be accepted.
	Measures	Changes the JOG mode after the JOG operation.
H2031 ＊	Error message	"Illegal parameter(JOGTSJ,JOGJSP)"
	Cause	The parameter JOGTSJ, JOGJSP settings are not correct. [JOGPSP]、[JOGJSP] = (element 1, element 2) = (constant high, constant low)
	Measures	Set the set dimension to 5 or less.
H2040	Error message	Teaching position is not correct.
	Cause	Change the teaching position.
	Measures	The work coordinates were not able to be calculated from the teaching position. The cause is the following content. 1. Two points are the same position. 2. Three points line up on the straight line. Confirm and correct the teaching position
L2041	Error message	Can't calculate frame transformation coordinates.
	Cause	The position data that defines the coordinate system used with frame transformation are on the same point or arranged on a straight line so the coordinate system could not be calculated.
	Measures	Change the position data to appropriate positions so the coordinate system can be calculated.

Error No.	Error cause and measures	
L2042	Error message	Frame transformation coordinates are not set
	Cause	Frame transformation was attempted even though the coordinate system for frame transformation was not set.
	Measures	Set the coordinate system or cancel frame transformation. Check that the MvSpl command argument <Frame transformation> designation is correct.
L2050	Error message	CPU processing time exceeds
	Cause	Because the function made effective is many at the same time
	Measures	The CPU processing time exceeded the limiting value. Please take measures of either of following. 1. If interference avoidance function is activated, changes some cylinder models into the sphere model, or reduces the number of the models for checking . 2. Invalidate some functions of following. Interference avoidance, User definition area, Free plane limit, Collision detection, Compliance, Tracking, Force sense 3. Reduces the Interrupt definition function currently used by the program.
H2090	Error message	In interference zone n. (n indicates the zone number (1 to 32).)
	Cause	Movement outside the user-defined area "n" range was attempted.
	Measures	Adjust the position.
H211n (n indicates the plane number (1 to 8).)	Error message	Free plane n overrun
	Cause	Movement outside the plane defined with free plane "n" was attempted.
	Measures	Adjust the position.
H212n (n indicates the plane number (1 to 8).)	Error message	Free plane n overrun
	Cause	Movement outside the plane defined with free plane "n" was attempted.
	Measures	Adjust the position.
H2129	Error message	Free plane limit data illegal
	Cause	The free plane data setting value is illegal.
	Measures	The two points are the same in the three points of parameter: SFCnp. The value of parameter: SFCnAT is except "0, 1, -1".
H213n (n indicates the axis number (1 to 8).)	Error message	Jn Speed is excessive (command)
	Cause	The speed instruction value of n axis exceeded the permissible value. When moving by the linear interpolation (or circle interpolation), needs to rotate the motor at the big speed depending on the pose. If the specified speed is large, the speed instruction value to the motor may exceed the permissible value.
	Measures	Please lower moving speed by the Ovrd command and Spd command or change the movement position.
H214n (n indicates the axis number (1 to 8).)	Error message	Jn +ABS limit over
	Cause	The axis "n" + ABS limit was exceeded.
	Measures	Referring to "Operation to Temporarily Reset an Error that Cannot Be Canceled" in the separate volume, "Instruction Manual/Detailed Explanation of Functions and Operations", reset the error and move the axis "n" within the operation range using JOG operation.
H215n (n indicates the axis number (1 to 8).)	Error message	Jn -ABS limit over
	Cause	The axis "n" - ABS limit was exceeded.
	Measures	Referring to "Operation to Temporarily Reset an Error that Cannot Be Canceled" in the separate volume, "Instruction Manual/Detailed Explanation of Functions and Operations", reset the error and move the axis "n" within the operation range using JOG operation.
H216n (n indicates the axis number (1 to 8).)	Error message	Jn (+) angle exceeds the limit
	Cause	The axis "n" + Joint limit was exceeded. When the teach mode, numerical "0" is set to n.
	Measures	Adjust the position.
H217n (n indicates the axis number (1 to 8).)	Error message	Jn (-) angle exceeds the limit
	Cause	The axis "n" - Joint limit was exceeded. When the teach mode, numerical "0" is set to n.
	Measures	Adjust the position.
H2181	Error message	X(+) data exceeds the limit
	Cause	The X axis + XYZ limit was exceeded.
	Measures	Adjust the position.

Error No.	Error cause and measures	
H2182	Error message	Y(+) data exceeds the limit
	Cause	The Yaxis + XYZ limit was exceeded.
	Measures	Adjust the position.
H2183	Error message	Z(+) data exceeds the limit
	Cause	The Z axis + XYZ limit was exceeded.
	Measures	Adjust the position.
H2191	Error message	X(−) data exceeds the limit
	Cause	X(−) data exceeds the limit
	Measures	Adjust the position.
H2192	Error message	Y(−) data exceeds the limit
	Cause	Y(−) data exceeds the limit
	Measures	Adjust the position.
H2193	Error message	Z(−) data exceeds the limit
	Cause	Z(−) data exceeds the limit
	Measures	Adjust the position.
H220m (m indicates the monitoring plane number (1 to 8).)	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	SLP (ROBOT position error:CMD Plane)
	Cause	SLP detected the robot position CMD over the monitoring plane
	Measures	Please confirm the robot movement or related parameter setting
	Error message	SLP (ROBOT position error:FB Plane)
	Cause	SLP detected the robot position FB over the monitoring plane
	Measures	Please confirm the robot movement or related parameter setting
	Error message	SLP (ROBOT position error:CMD Area)
	Cause	SLP detected the robot position CMD over the monitoring area
	Measures	Please confirm the robot movement or related parameter setting
	Error message	SLP (ROBOT position error:FB Area)
	Cause	SLP detected the robot position FB over the monitoring area
	Measures	Please confirm the robot movement or related parameter setting
H221n (n indicates the axis number (1 to 8).)	Error message	STR (ROBOT torque error)
	Cause	Safety Torque Range monitoring detected over the torque limit
	Measures	A feedback torque in excess of the predetermined allowable torque width is detected by the torque width monitoring function.
H222m * (m indicates the DSI number (1 to 8).)	Error message	DSI inconsistency
	Cause	The duplex DSI status is not consistent.
	Measures	The duplex DSI signal ON/OFF status is not consistent. Check the following for the DSI. ・ DSI wiring ・ Duplex signal ON/OFF status ・ ON/OFF status switching timing (The error occurs when the ON/OFF status remains inconsistent between the duplex signals for about 0.1 seconds or more.)
H2230 *	Error message	Mismatch of Dual Safety Output
	Cause	State of the redundant wiring do not match (Dual Safty Output)
	Measures	Turn the power OFF and ON once If the same error recurs, contact the manufacturer.
H2231 *	Error message	Mismatch of Dual Safety Output Feedback
	Cause	State of Dual Safty Output and the feeedback do not match
	Measures	Turn the power OFF and ON once If the same error recurs, contact the manufacturer.
H2240 *	Error message	Origin data change
	Cause	The origin data is changed during execution of the position monitoring function.
	Measures	The error occurs when the origin data is changed by setting the origin during execution of the position monitoring function. After turning OFF the power, turn ON the power again to reset the error.

Error No.	Error cause and measures	
C2250	Error message	Safety function execution disabled (No origin setting)
	Cause	The safety function cannot be executed because the origin is not set.
	Measures	The safety monitoring function is not activated when the origin is not set. Set the origin.
H2260 *	Error message	Safety function execution disabled (No extended safety unit)
	Cause	The extended safety unit of the robot safety option is not connected.
	Measures	To use the safety monitoring function, it is necessary to connect the extended safety unit of the robot safety option to the controller. Connect the extended safety unit. If the error occurs even if the extended safety unit is connected, the extended safety unit may be faulty. Contact your service provider.
H2261 *	Error message	Extended safety unit's No. error
	Cause	The extended safety unit' s station No. is illegal
	Measures	Set the extended safety unit' s station No. to 2
H2270	Error message	Cannot servo ON (SS1/STO active)
	Cause	The servo cannot be turned ON while SS1/STO is active
	Measures	Disable SS1 before turning the servo ON
H2280	Error message	SS1 deceleration time exceeded
	Cause	The robot didn't stop within deceleration time from SS1 enabled
	Measures	Please confirm the robot movement and the load, stop speed parameter (SFSPZERO) setting
H2281	Error message	SS2 deceleration time exceeded
	Cause	The robot didn't stop within deceleration time from SS2 enabled
	Measures	Please confirm the robot movement and the load, stop speed parameter (SFSPZERO) setting
H2282	Error message	SOS (Position error)
	Cause	Detect the change of position FB on SOS
	Measures	An error that the position FB has changed on SOS due to external force was detected. Eliminate the cause of the external force or perform the risk assessment. Then modify the value of the parameter "SOSTLRNC" to extend the acceptable range of SOS.
	Error message	SOS (Speed error)
	Cause	Detect FB speed over on SOS
	Measures	Please check external force or related parameter settings
	Error message	SOS (Position command error)
	Cause	Detect the change of position CMD on SOS.
	Measures	Please confirm the robot movement or related parameter setting
	Error message	SOS (Speed command error)
	Cause	Detect CMD speed over on SOS.
	Measures	Please confirm the robot movement or related parameter setting
H230n (n indicates the axis number (1 to 8).)	Error message	SLS (Joint Speed Error)
	Cause	Speed monitor detected the speed over
	Measures	A speed feedback exceeding the limit speed is detected by the speed monitoring function. Check the robot movement or the monitoring speed setting. Or else, check the delay time (parameterSLSDLY) setting.
H231n (n indicates the direction in which an error is detected, 1: Composite speed, 2: X+, 3: X−, 4: Y+, 5: Y−, 6: Z+, 7: Z−)	Error message	SLS (XYZ Speed Error)
	Cause	Speed monitor detected the speed over
	Measures	A speed feedback exceeding the limit speed is detected by the speed monitoring function. Check the robot movement or the monitoring speed setting. Or else, check the delay time (parameterSLSDLY) setting.
H2320	Error message	SF robot control error
	Cause	The robot motion command and the feedback are inconsistent.
	Measures	The robot position command and the feedback position are inconsistent. Check the details of the robot movement, the terminal load setting, or interference with the peripheral equipment.

Error No.	Error cause and measures	
H2370 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	SF (Process error)
	Cause	The Safety Function is not normally executed
	Measures	Turn the power OFF and ON once. If the same error recurs, contact the manufacturer.
	Error message	SF (Process Counter error)
	Cause	The Safety Function of servo CPU is not normally executed
	Measures	Turn the power OFF and ON once. If the same error recurs, contact the manufacturer.
L240n (n: Robot CPU No.)	Error message	Collision avoidance detect (*) Note) "(*)": shows the detected model. (the the lower digit of the parameter for the model registry)
	Cause	A collision avoidance was detected
	Measures	Release the collision avoidance state
L241n (n: Robot CPU No.)	Error message	Collision avoidance detect (*) Note) "(*)": shows the detected model. (the the lower digit of the parameter for the model registry)
	Cause	A collision avoidance was detected
	Measures	Release the collision avoidance state
L2420	Error message	Collision avoidance comm. error
	Cause	Collision avoidance comm. error
	Measures	Check the robot controller
L2421	Error message	A lot of colli. avoidance models
	Cause	A lot of collision avoidance models
	Measures	Reduce the collision avoidance models
L2430	Error message	Collision avoidance re-detect
	Cause	Collision avoidance was already detected
	Measures	Move the robot arm from the interference area and resume the operation.
L2500	Error message	Tracking encoder data error
	Cause	An error was detected in the data of tracking encoder.
	Measures	1) The conveyor rotates at the fixed velocity. 2) The connection of the encoder. 3) The earth of the earth wire.
L2510	Error message	Tracking parameter reverses
	Cause	Tracking parameter[EXCRGM*] Setting value reverses
	Measures	Check the parameter[EXCRGM*] value.
L2520	Error message	Tracking parameter is range over
	Cause	Tracking parameter[TRBUF] Setting value is range over. Setting range: element 1=1-8, element 2=1-64
	Measures	Check the parameter[TRUBF] value.
L2530	Error message	There is no area where data is written.
	Cause	There is no area where data is written.
	Measures	Please read the data by using TrRd
L2540	Error message	There is no read data.
	Cause	There is no read data.
	Measures	Please use TrRd after executing TrWrt.
L2560	Error message	Ilegal parameter of Tracking.
	Cause	The value of the parameter: EXTENC is outside the range. setting range: 1-8.
	Measures	Please check the value of parameter.
L2580	Error message	No workpiece in the tracking area.
	Cause	There is no workpiece in the tracking buffer or "TrkMv On" command is executed. Before the workpiece enters to the tracking area.
	Measures	Execute "TrkMv On" command when the workpiece is in the tracking area.

Error No.	Error cause and measures	
L2601	Error message	Start pos. exseeds the limit
	Cause	The start position is outside the operation range.
	Measures	Adjust the position.
L2602	Error message	DSTN pos. exseeds the limit
	Cause	The target position is outside the operation range.
	Measures	Adjust the position.
L2603	Error message	Med pos. data exseeds the limit
	Cause	The intermediate position is outside the operation range.
	Measures	Adjust the position.
L2610	Error message	Spline interpolation error (Spline file)
	Cause	An error related to the spline file occurred.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	Can't open spline file
	Cause	The spline file corresponding to the spline No. designated with the MvSpl command could not be opened.
	Measures	Check that the spline No. designation is correct, and that the target spline file is saved in the controller.
	Error message	Spline file is broken
	Cause	The spline file contents did not match the checksum.
	Measures	Open the target spline file and check the contents. Then, save the file again to recreate the spline file. Write the new file to the controller again.
	Error message	Spline file is not supported.
	Cause	The designated spline file cannot be used with the current controller.
	Measures	Check the spline file and controller versions. It may be necessary to upgrade the software version.
	Error message	Can't change spline file.
	Cause	The target spline file is currently being used for spline interpolation (file is open).
	Measures	A spline file currently being used for spline interpolation (file is open) cannot be exported to the controller, deleted or renamed. Carry these out after spline interpolation ends.
	Error message	Can't get data.
	Cause	The spline file is closed so the data cannot be retrieved.
	Measures	Reset the program, and then execute the MvSpl command again.
	Error message	The setting of Ex-T is illegal
	Cause	The setting of Ex-T is illegal
	Measures	Please confirm the number and the content of the file
	Error message	Cannot make the spline file
	Cause	The specified spline file is opened already
	Measures	Please confirm the specified number

Error No.	Error cause and measures	
L2611	Error message	Spline interpolation error (path point)
	Cause	An error related to the path point registered in the spline file has occurred.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	Path points are too close (nnnn)
	Cause	The distance between path points is too short, or the speed command in respect to the path point distance is too high.
	Measures	"nnnn" in the error message indicates the path point No. causing the error. Review this path point's position, or review the spline interpolation command speed. * The high speed spline interpolation command has been added to the software version A3a. With this function enabled, a command speed higher than what was previously possible can be specified for the distance between the respective path points. Refer to the separate manual "Instruction Manual/Detailed Explanation of Functions and Operations" in details.
	Error message	The posture variation is too large (nnnn)
	Cause	The variation in posture between path points is too large. (The posture variation angle exceeds 150 degrees.)
	Measures	"nnnn" in the error message indicates the path point No. causing the error. Review the posture so that the posture variation amount for this path point is smaller, or add a path point to reduce the posture variation amount in the single block.
	Error message	Path point's configuration flag is incorrect (nnnn)
	Cause	A different path point is registered for the configuration flag value.
	Measures	"nnnn" in the error message indicates the path point No. causing the error. Review this path point's position, and change it so it is the same configuration flag as the other path points.
	Error message	Path point is not registered
	Cause	The Specified path point is not registered
	Measures	Please confirm the number and the content of the file
	Error message	Cannot open the path point file
	Cause	The specified path point file does not exist
	Measures	Please confirm the file name and existence of the file
	Error message	The format of file is different
	Cause	The format of file is different
	Measures	Please confirm the format of the file
	Error message	The number of point is illegal
	Cause	The number of path point is illegal
	Measures	Please confirm the file and the content of the file

Error No.	Error cause and measures	
L2612	Error message	Spline interpolation error (Execution error)
	Cause	An error related to the spline interpolation execution conditions occurred.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	This robot does not support spline interpolation.
	Cause	Spline interpolation was attempted with a robot that does not support spline interpolation.
	Measures	Use a different movement command than spline interpolation.
	Error message	Can't execute with these start conditions.
	Cause	Spline interpolation was attempted with a robot program in a slot having the start conditions ALWAYS·ERROR.
	Measures	Spline interpolation cannot be executed with a slot having the start conditions ALWAYS·ERROR. Delete spline interpolation or change the start conditions to START.
	Error message	Another spline interpolation is being executed
	Cause	Multiple spline interpolations cannot be executed simultaneously.
	Measures	Check whether spline interpolation was attempted with a different robot during spline interpolation, or whether spline interpolation was directly attempted while spline interpolation was halted.
	Error message	Can't execute step return
	Cause	Step return was attempted in respect to spline interpolation.
	Measures	Spline interpolation does not support the step return operation. Do not attempt step return.

Error No.	Error cause and measures	
L2613	Error message	Spline interpolation error (interpolation process)
	Cause	An error occurred during the spline interpolation process.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	Not enough path points
	Cause	There are less than four path points registered in the spline file.
	Measures	At least four path points are required to execute spline interpolation. Use a spline file in which four or more path points are registered.
	Error message	Not enough arc designated points (nnnn)
	Cause	Three consecutive path points are not designated for the spline interpolation arc.
	Measures	The number of points in the arc containing the path point No. "nnnn" displayed in the error message is insufficient. To generate an arc, an arc for three consecutive path points must be designated. This error occurs if there are only two consecutive points. Add another path point in the arc designation to move with an arc path.
	Error message	Block data calculation error (ssssssss)
	Cause	An error occurred in the process to calculate the data related to the spline interpolation block.
	Measures	The details of the error differ according to "ssssssss" in the error message. ・Reg.Pt.: The information on the number of path points saved in the spline file does not match the number of path points actually registered. Open the spline file in the Spline File Edit screen once, save it again and export it to the controller. ・Cir.Arc: The arc could not be generated. Check that the path points for the arc designation are not arranged on a straight line. ・Frm.Cnv.: The frame transformation calculation failed. Review the path point positions. ・Blk.Stp.: Generation of the data for the block stopped, and spline interpolation could not be executed. Reset the program. ・PtoJ.: The joint angle cannot be calculated at the position. Review the path point positions.
	Error message	Block data does not exist
	Cause	The load in the process for the controller during spline interpolation was large, and the data for the spline interpolation block could not be generated in time.
	Measures	Check whether the load in the spline interpolation execution process can be reduced in the following ways. ・Stop simultaneous execution of functions related to movement such as the collision detection function or visual control function. ・Review the multi-tasks and reduce the number of slots executed simultaneously. ・Reduce the dedicated output signal assignments. ・Reduce the spline interpolation command speed.
	Error message	Spline interpolation command calculation error (nnnn)
	Cause	An error occurred in the process for calculating the spline interpolation position commands.
	Measures	Review the position of path point No. "nnnn" indicated in the error message or the MvSpl command argument's setting value. Check that the cancel angle is set correctly. (Does the spline curve bend suddenly?)

Error No.	Error cause and measures	
L2614	Error message	Spline interpolation error (other functions)
	Cause	A function that cannot be used with the spline interpolation was executed.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	Tracking function is enabled
	Cause	The tracking function was enabled when attempting spline interpolation.
	Measures	Spline interpolation and tracking function cannot be executed simultaneously. Disable the tracking function before executing spline interpolation.
	Error message	Can't change tool/base setting
	Cause	The tool/base setting was changed during spline interpolation execution.
	Measures	The tool/base setting cannot be changed during spline interpolation (including when halted). Change the settings after spline interpolation ends.
	Error message	Can't execute Jrc command
	Cause	The Jrc command was executed during spline interpolation.
	Measures	The Jrc command cannot be executed during spline interpolation (including when halted). Execute the command after spline interpolation ends.
L2615	Error message	Spline interpolation error (exceeds setting range)
	Cause	A setting value related to spline interpolation exceeds the setting range.
	Measures	Refer to the error details No. and check the details of the occurring error. Then take actions for those error details.
	Error message	M_SplVar setting value exceeds setting range
	Cause	A value exceeding the range was substituted into M_SplVar.
	Measures	Substitute a value within the setting range (0 to 32767).
	Error message	M_SplVar setting range exceeds setting range (nnnn)
	Cause	A value exceeding the setting range is set in the spline file.
	Measures	Change the setting for the path point No. "nnnn" indicated in the error message so it is within the range (−1 to 32767).
	Error message	Tolerance setting value exceeds range (nnnn)
	Cause	The tolerance designation in the spline file exceeds the range.
	Measures	Change the tolerance designation for the path point No. "nnnn" indicated in the error message so it is within the range (0 to 100).
	Error message	Output signal exceeds range (nnnn)
	Cause	The head No. for the signal output in the spline file exceeds the range.
	Measures	Change the head No. of the signal output for the path point No. "nnnn" indicated in the error message so it is within the range (−1 to 32767).
	Error message	Interpolation setting information exceeds range
	Cause	Data with value exceeding the range was found in the spline file interpolation setting information.
	Measures	Open the target spline file and check the contents. Then, save the file again to recreate the spline file. Write the new file to the controller again.
	Error message	Header information exceeds range
	Cause	Data with value exceeding the range was found in the spline file header information.
	Measures	Open the target spline file and check the contents. Then, save the file again to recreate the spline file. Write the new file to the controller again.
L2621	Error message	Tracking function is enabled.
	Cause	The tracking function and Ex-T control function cannot be enabled simultaneously.
	Measures	When using the Ex-T control function, disable the tracking function.
L2622	Error message	Singular point passage function is enabled.
	Cause	The singular point passage function and Ex-T control function cannot be enabled simultaneously.
	Measures	When using the Ex-T control function, disable the singular point passage function.

Error No.	Error cause and measures	
H264n (n indicates the axis number (1 to 8))	Error message	Add axis FLS signal is input
	Cause	Add axis FLS signal is input
	Measures	Release the brake and move the arm by hand. Or, referring to "Operation to Temporarily Reset an Error that Cannot Be Canceled" in the separate volume "Instruction Manual/Detailed Explanation of Functions and Operations", reset the error and move the axis "n" within the operation range using JOG operation.
H265n (n indicates the axis number (1 to 8))	Error message	Add axis RLS signal is input
	Cause	Add axis RLS signal is input
	Measures	Release the brake and move the arm by hand. Or, referring to "Operation to Temporarily Reset an Error that Cannot Be Canceled" in the separate volume "Instruction Manual/Detailed Explanation of Functions and Operations", reset the error and move the axis "n" within the operation range using JOG operation.
L2660	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Error concerning axis coop
	Cause	The error concerning the axis coop occurred
	Measures	Please Confirm the content by a detail number of the error
	Error message	Illegal robot No.(axis trk)
	Cause	The designated robot No. is illegal
	Measures	Set the correct robot No.
	Error message	Illegal axis No.(axis trk)
	Cause	The designated axis No. is illegal
	Measures	Set the correct axis No.
	Error message	Illegal unit system (axis trk)
	Cause	A rotary axis is designated
	Measures	Please designate a linear drive axis
	Error message	Origin unsetting (axis trk)
	Cause	Axis tracking cannot be executed because of origin unsetting
	Measures	Please set the origin and turn the power OFF and ON
	Error message	Illegal robot No.(base coop)
	Cause	The designated robot No. is illegal
	Measures	Set the correct robot No.
	Error message	Illegal axis No.(base coop)
	Cause	The designated axis No. is illegal
	Measures	Set the correct axis No.
	Error message	Illegal unit system (base coop)
	Cause	A rotary axis is designated
	Measures	Please designate a linear drive axis
	Error message	Origin unsetting (base coop)
	Cause	Base coop cannot be executed because of origin unsetting
	Measures	Please set the origin and turn the power OFF and ON

Error No.	Error cause and measures	
L2661	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Error concerning axis coop (combi.)
	Cause	The function cannot be used at the same time with the axis coop
	Measures	Please confirm the content by a detailed number of the error
	Error message	Cannot be used (axis trk)
	Cause	A synchronous addition axis control is effective
	Measures	Invalidate a synchronous addition axis control
	Error message	Cannot be used (base coop)
	Cause	Tracking function is effective
	Measures	Invalidate the tracking function
	Error message	Cannot be used (base coop)
	Cause	A synchronous addition axis control is effective
	Measures	Invalidate a synchronous addition axis control
	Error message	Jrc cannot be executed
	Cause	The base coop is executing
	Measures	Please do not use Jrc command
	Error message	Interpolation cannot be executed
	Cause	An addition axis is going to move
	Measures	Please do not move an addition axis
	Error message	Cannot be used (base coop)
	Cause	Interference avoidance function is effective
	Measures	Invalidate the interference avoidance function
L2662	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Work setting cannot be changed
	Cause	The additional axis tracking is executing
	Measures	Please change after the work coop is invalid
	Error message	This work cannot use
	Cause	The setting is not done(axis trk)
	Measures	Please confirm a set valuse and then execute
H2663	Error message	Origin data was changed
	Cause	Origin data was changed about the base cooperation target axis
	Measures	Turn the power OFF and ON once
L2700	Error message	Cmp error (different mode)
	Cause	The designated mode is different from the current mode.
	Measures	Execute Cmp Off and then designate.
C2710	Error message	Cmp error (displacement)
	Cause	The displacement magnitude of the compliance operation exceeded the specified value.
	Measures	Correct the program, position or other item so that the displacement magnitude can be reduced.
H2720	Error message	Cmp error (joint angle)
	Cause	Cmp Command exceeds the limit of a joint angle.
	Measures	Adjust the position data or reduce the displacement.
C272n (n indicates the axis number (1 to 8).)	Error message	Cmp error (Jn joint angle)
	Cause	Cmp Command exceeds the limit of joint angle of joint "n" axis
	Measures	Change the position data or reduce displacement.
C273n (n indicates the axis number (1 to 8).)	Error message	Cmp error (Jn axis speed)
	Cause	Cmp Command exceeds the limit of speed of joint "n" axis.
	Measures	Change the position data or slow down.

Error No.	Error cause and measures	
C2740	Error message	Cmp error (coordinates conv.)
	Cause	An error was detected in the coordinates conversion of Cmp command.
	Measures	Adjust the position data.
L2750	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Cannot execute while tracking
	Cause	Cannot execute while tracking
	Measures	Execute Trk Off and then execute
	Error message	Cannot invalidate F.Ctrl (Trk)
	Cause	Force control cannot be invalidated while executing tracking
	Measures	Please invalidate the tracking function
	Error message	Unable to disable force sense control. (Tracking being performed)
	Cause	Unable to disable force sense control while tracking function being executed.
	Measures	Try again after disabling the tracking function.
H2760	Error message	The force sense control offset limit was reached.
	Cause	The robot attempted to move beyond the force sense control offset limit.
	Measures	Check whether there is a problem with robot movement while force sense control is enabled. (The offset limit is the value set in parameter FSCORMX.)
H2770	Error message	Outside offset position movement range (**) Note) "**" is substituted with "±Jn" (n is axis No.)
	Cause	The position after force sense control offset lies outside the range. The robot may have been moved near the movement range limit.
	Measures	Review the movement position or force sense control settings, and ensure that the offset position does not exceed the movement range.
H2780	Error message	Offset position speed over (**) Note) "**" is substituted with "Jn" (n is axis No.)
	Cause	The speed of movement to the position after offsetting with force sense control exceeded the speed limit. The movement speed may be too fast, or the robot may have been moved at the singular point adjacent.
	Measures	Review the movement speed and movement position, or the force sense control settings.
L2800	Error message	Illegal position data
	Cause	This may occur for a position to which the robot cannot reach.
	Measures	Adjust the position.
L2801	Error message	Illegal position data (start)
	Cause	This may occur for a starting position to which the robot cannot reach.
	Measures	Adjust the position.
L2802	Error message	Illegal position data (dstn)
	Cause	This may occur for a ending position to which the robot cannot reach.
	Measures	Adjust the position.
L2803	Error message	Illegal assisting position data (intmed)
	Cause	The intermediate path at the straight line interpolation and the route at circle interpolation are the position which the robot cannot move.
	Measures	Confirm the error occurrence line and confirm that there is no position which the robot cannot move. And please correct the data of starting position, midway position, or ending position .
L2810	Error message	Posture flag is disagree
	Cause	The structure flag of the start point and end point don't match.
	Measures	Adjust the position data.
H2820	Error message	Illegal Accel ratio
	Cause	This occurs when the acceleration/deceleration ratio is too small.
	Measures	Adjust the acceleration/deceleration ratio to a larger value.
H2830	Error message	System error (ipol posture type)
	Cause	The Type argument of the Mov instruction was set to −1 or a similar value.
	Measures	Change the Type argument of the Mov instruction to a correct value (0, 1, etc).

Error No.	Error cause and measures	
H2840	Error message	System error (ipol parameters)
	Cause	The parameter may have been damaged.
	Measures	If it comes back, contact your service provider.
H2850	Error message	System error (ipol norm)
	Cause	The norm is illegal. A problem occurred in internal computation processing.
	Measures	If it comes back, contact your service provider.
H2860	Error message	System error (ipol type)
	Cause	An illegal interpolation process method is being used. A problem occurred in internal computation processing.
	Measures	If it comes back, contact your service provider.
H2870	Error message	System error (ipol data undef)
	Cause	The interpolation position data has not been defined. A problem occurred in internal computation processing.
	Measures	If it comes back, contact your service provider.
H2880 *	Error message	System error (ipol data area)
	Cause	The memory is insufficient for the inside operation.
	Measures	If it comes back, contact your service provider.
H2890	Error message	System error (undefined err)
	Cause	An undefined error number was generated in internal computation processing.
	Measures	If it comes back, contact your service provider.
L2900	Error message	System ERROR M00 to M04
	Cause	An error occurred in the internal processing.
	Measures	If it comes back, contact your service provider.
L3100	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Processor Call Stack over
	Cause	Too many function calls or local variables.
	Measures	Correct the program and re-execute.
	Error message	System ERROR (Call Stack)
	Cause	Processor Call Stack defect
	Measures	Possible problem with execution sequence of Function command.

Error No.	Error cause and measures	
L3110	One of the errors below is detected. Please take measures corresponding to an error message. Refer to the separate manual, “Instruction Manual/Detailed Explanation of Functions and Operations” or “Instruction Manual/Force Sensor Function” in details.	
	Error message	Arg. value range over
	Cause	Arg. value is outside range. For the PtoJ function, a position which the robot cannot reach or a singular point was set as the arg. value.
	Measures	Please confirm the range of the argument and set a correct value
	Error message	The force sense control command argument lies outside the range.
	Cause	A value outside the range was set for the force sense control command argument.
	Measures	Check the argument range and set a correct value.
	Error message	The force sense control status variable argument lies outside the range.
	Cause	A value outside the range was set for the force sense control status variable argument.
	Measures	Check the argument range and set a correct value.
	Error message	The force sense control related argument lies outside the range.
	Cause	A value outside the range was set for the force sense control related argument.
	Measures	Check the argument range and set a correct value.
	Error message	The Mo trigger No. lies outside the range.
	Cause	A value outside the range was set for the Mo trigger No.
	Measures	Check the setting range and set a correct value.
	Error message	Def MoTrg command argument error
	Cause	An unusable variable or different mechanical No. was set.
	Measures	Set a usable variable or same mechanical No.
	Error message	Arg. value range over (MVA2)
	Cause	Argument value range is over (MVA2)
	Measures	Check the argument range and reinput
L3120	Error message	No. of arg. is over
	Cause	No. of argument is over
	Measures	Please confirm the range of the argument and set a correct value
L3130	Error message	COM file is already opened
	Cause	Opening of a file already opened was attempted.
	Measures	Check the file No. and re-execute.
L3140	Error message	Can't open COM file
	Cause	The file cannot be opened.
	Measures	Check the file No. and re-execute.
L3141	Error message	The NVOpen command is not executed.
	Cause	No NVOpen command was executed before execution of a command communicating with the vision sensor.
	Measures	Revise the robot program to execute the NVOpen command.
L3142	Error message	The communication line can not be opened.
	Cause	The line for communication with the vision sensor can not be opened.
	Measures	Check the communication cable or the communications parameters.
L3150	Error message	Cannot Print (INPUT mode)
	Cause	The file open mode is INPUT, so writing is not possible.
	Measures	Check the file No. and open mode, and re-execute.
L3170	Error message	Cannot Input (OUTPUT mode)
	Cause	The file open mode is OUTPUT, so writing is not possible.
	Measures	Check the file No. and open mode, and re-execute.
L3180	Error message	System error (array range over)
	Cause	System error (array range over)
	Measures	If it comes back, contact your service provider

Error No.	Error cause and measures	
L3200	Error message	This file is read only
	Cause	The file cannot be read.
	Measures	Check the contents of the file.
L3210	Error message	This variable is write protected
	Cause	Writing of this variable is prohibited.
	Measures	Check the variable protection setting.
L3220	Error message	Nesting over
	Cause	A nest-over error occurred in If of the If instruction or For of the For instruction.
	Measures	Correct the program and re-execute.
L3230	Error message	For Next statements unmatch
	Cause	The No. of For and Next statements do not match.
	Measures	Correct the program and re-execute.
L3240	Error message	Nesting over (For, While)
	Cause	Nesting over (For, While)
	Measures	Correct the program and re-execute.
L3250	Error message	While WEnd statements unmatch
	Cause	The No. of While and WEnd statements do not match.
	Measures	Correct the program and re-execute.
L3251	Error message	Number of jump destination exceeds 32
	Cause	The number of branches defined exceeded 32.
	Measures	Correct the program and re-execute.
L3252	Error message	If EndIf statements unmatch
	Cause	If EndIf statements unmatch.
	Measures	Correct the program and re-execute.
L3253	Error message	Nesting over (If, EndIf)
	Cause	Nesting over (If, EndIf)
	Measures	Correct the program and re-execute.
L3254	Error message	Select – End Select statements unmatch
	Cause	Select – End Select statements unmatch.
	Measures	Correct the program and re-execute.
L3255	Error message	If Else statements unmatch
	Cause	If Else statements unmatch.
	Measures	Correct the program and re-execute.

Error No.	Error cause and measures	
L3256	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Function definition error
	Cause	The error concerning the function definition occurred.
	Measures	Please confirm the content by a detailed number of the error.
	Error message	Function definitions exceeds max.
	Cause	The number of function definitions exceeds 256.
	Measures	Correct the program and re-execute.
	Error message	Function statement is incorrect
	Cause	Function statement position is incorrect.
	Measures	Correct the program and re-execute.
	Error message	Function FEnd statemets unmatch
	Cause	Function FEnd statemets unmatch.
	Measures	Correct the program and re-execute.
	Error message	Undefined function
	Cause	The called function or the assigned function is undefined.
	Measures	Please confirm the function procedure name.
	Error message	Function Main called
	Cause	Function Main called.
	Measures	Function Main cannot be called.
	Error message	Function Main does not exist
	Cause	Programs that there is no Function Main cannot be executed.
	Measures	Please add Function Main (parameterless) to the program.
	Error message	Local var. in function was used
	Cause	A local variable in the function procedure was used.
	Measures	Local variables in a function procedure cannot be used.
L3257	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Include declaration error
	Cause	The error concerning the include declaration occurred.
	Measures	Please confirm the content by a detailed number of the error.
	Error message	Include declaration exceeds max.
	Cause	The number of include declarations exceeds 64.
	Measures	Correct the program and re-execute.
	Error message	Include statement is incorrect
	Cause	Include statement position is incorrect.
	Measures	Correct the program and re-execute.
L3258	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Preprocessor error
	Cause	The error concerning the preprocessor occurred.
	Measures	Please confirm the content by a detailed number of the error.
	Error message	Preprocessor run line exceeds max.
	Cause	Preprocessor execution line exceeds 1000 lines.
	Measures	Correct the program and re-execute.
	Error message	Preprocessor run error
	Cause	Commands that can not execute by the preprocessor are included.
	Measures	Correct the program and re-execute.

Error No.	Error cause and measures	
L3260	Error message	Cannot exec for all slots
	Cause	Execution with all slots designated is not possible.
	Measures	Designate an individual slot and try again.
L3270	Error message	The command size is exceeded
	Cause	The command size is exceeded.
	Measures	Specify within single-byte 256 characters.
L3280	Error message	Cannot execute without GetM
	Cause	The command you attempted to execute cannot be executed without GetM. Or, a non-existing mechanical number was specified.
	Measures	Execute it after executing the RelM and GetM commands in another task slot.
L3281	Error message	Cannot execute during RUN
	Cause	Cannot execute during operation.
	Measures	Cannot execute during operation.
L3282	Error message	"Can't RUN (not select, attribute)"
	Cause	The program is not selected or the attribute is illegal.
	Measures	Load the program into the specified task slot. Or, change the program attributes.
L3285	Error message	Cannot execute (RUN or WAI)
	Cause	can't execute in the state of stopping or executing.
	Measures	Reset the program (cancel the abort status).
L3286	Error message	Program is empty
	Cause	Execution of an empty program was attempted.
	Measures	Make the program or select the correct program.
L3287	Error message	Cannot execute (ERROR ALWAYS)
	Cause	This command cannot be used when the start conditions are ERROR and ALWAYS.
	Measures	Correct the program.
L3288	Error message	Cannot execute while editing
	Cause	That program cannot be executed because it is being edited.
	Measures	Finish editing the program first, and then start it.
L3289	Error message	Program does not exist (SLT*)
	Cause	The program designated in the slot table does not exist.
	Measures	Correct the slot parameter.
L3290	Error message	System slot cannot be executed
	Cause	The system slot cannot be executed.
	Measures	Check whether another slot (user slot) is being operated.
L3300	Error message	User slot cannot be executed
	Cause	A user slot cannot be executed.
	Measures	Check whether the system slot is being operated.
L3310	Error message	Cannot execute XRun (Runing)
	Cause	XRun is not possible as the designated slot is operating.
	Measures	Stop the specification slot, and execute.
L3320	Error message	Cannnot execute XRun (empty)
	Cause	XRun is not possible as the program has not been selected.
	Measures	Specify program name to the argument or execute XLoad.
L3330	Error message	Cannnot execute XStp (empty)
	Cause	XStp is not possible as the program has not been selected.
	Measures	correct the program, and execute.
L3340	Error message	Cannnot execute XRst (empty)
	Cause	XRst is not possible as the program has not been selected.
	Measures	Resetting is possible in the state of waiting only.

Error No.	Error cause and measures	
L3350	Error message	Cannnot execute XRst (Running)
	Cause	Xrst is not possible as the program is executing.
	Measures	Stop execution, and do it.
L3360	Error message	Cannnot execute XLoad (not PSA)
	Cause	XLoad cannot be executed when the program cannot be selected.
	Measures	Execute XRst, and do it.
L3361	Error message	Can not load the program (SLT*)
	Cause	A non-existing program was specified in the slot parameter (SLTn).
	Measures	A non-existing program was specified in the slot parameter (SLTn).
L3370	Error message	Cannnot execute XClr (empty)
	Cause	XClr is not possible as the program has not been selected.
	Measures	XClr can only be executed to enable program selection.
L3380	Error message	Cannnot execute XClr (not PSA)
	Cause	Program selection is not enabled.
	Measures	Execute XClr after resetting the program (canceling the abort status).
L3390	Error message	Cannot use arc pallet
	Cause	Cannot use arc pallet.
	Measures	Change to another method.
L3391	Error message	Cannot use MVA2
	Cause	Cannot use MVA2
	Measures	Change to another method
L3400	Error message	System error (PROC stack over)
	Cause	System error. (Processor stack overflow)
	Measures	If it comes back, contact your service provider
L3500	Error message	Illegal format input (Input)
	Cause	The type of the variable specified by Input and the type of the received data do not match.
	Measures	Check the format.
L3501	Error message	Illegal Receive data (EBREAD)
	Cause	Type is different (receive data and specified variable)
	Measures	Please confirm specified tag data of the vision.
L3600	Error message	Jump destination does not exist
	Cause	No jump destination was found for the Def Act, On Com and On GoTo commands.
	Measures	Check the jump destination.
L3601	Error message	Jump destination is incorrect
	Cause	Jumping into program control block is prohibited.
	Measures	Correct the program and re-execute.
L3700	Error message	Undefined variable
	Cause	It was attempted to reference a variable that has not been initialized.
	Measures	Define a variable, enter an initial value, and then use it.
L3710	Error message	Nesting over (CallP)
	Cause	Program Call is used more than the limitation.
	Measures	Reduce the call count of CallP (nesting).
L3720	Error message	RC NX statements unmatch
	Cause	RC NX statements unmatch
	Measures	Match the numbers of RC and NX.
L3750	Error message	Illegal positions (Def Plt)
	Cause	Multi rotation flag (FL2 J1/J4 axis) is different
	Measures	Specify position to become the same multi rotation flag.

Error No.	Error cause and measures	
L3760	Error message	Illegal positions (Def Plt)
	Cause	J1 or J4 axis is greatly changed (Def Plt)
	Measures	Specify positions to be not different greatly
L3770	Error message	This is an undefined Mo trigger.
	Cause	An attempt was made to use an undefined Mo trigger.
	Measures	Define the specified Mo trigger before use.
L3780	Error message	Cannot use the MELFA Smart Plus.
	Cause	The MELFA Smart Plus card or MELFA Smart Plus card pack is not installed in the controller. When the MELFA Smart Plus card is installed, the setting of the parameter SMART+1 may be set incorrectly.
	Measures	Install the MELFA Smart Plus card or MELFA Smart Plus card pack. Set the parameter SMART+1 correctly.
L3781	Error message	Cannot use the MELFA Smart Plus.
	Cause	The MELFA Smart Plus card or MELFA Smart Plus card pack is not installed in the controller. When the MELFA Smart Plus card is installed, the setting of the parameter SMART+1 may be set incorrectly.
	Measures	Install the MELFA Smart Plus card or MELFA Smart Plus card pack. Set the parameter SMART+1 correctly.
L3810	Error message	Different argument type
	Cause	The type of an argument in an arithmetic operation, monadic operation, comparison operation or each function is different.
	Measures	Designate the correct argument.
L3820	Error message	Undefined intermediate code
	Cause	A program or system status variable may have been damaged.
	Measures	Restore using the backup data. If the backup data is not available, it is necessary to create a program again.
L3821	Error message	A compilation error occurred
	Cause	There is an error in the input syntax.
	Measures	Check the error number in the error message.
L3830	Error message	Cannot execute GetM
	Cause	GET of the mechanisms is not possible.
	Measures	Check whether the designated mechanisms are being used with a different slot.
L3840	Error message	Return without GoSub
	Cause	Return was executed without using GoSub.
	Measures	Check the program.
L3850	Error message	Undefined PLT
	Cause	The Def Plt command was not executed.
	Measures	Use it after defining a pallet with the Def Plt command.
L3860	Error message	Illegal position data defined
	Cause	There is an error in the position data.
	Measures	Check the position data definition.

Error No.	Error cause and measures	
L3870	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Illegal mecha No. (****) Note) "****" is substituted with the "robot status variable".
	Cause	The designated mecha No. is an invalid value
	Measures	Please set a correct mecha No.
	Error message	The force sense control status variable mechanical No. is an invalid value.
	Cause	An invalid variable was set for the force sense control status variable mechanical No.
	Measures	Set a correct mechanical No.
	Error message	The mechanical No. specified with the Def MoTrg command is an invalid value.
	Cause	The mechanical No. specified with the Def MoTrg command is an invalid value.
	Measures	Set a correct mechanical No.
L3880	Error message	Illegal slot No.
	Cause	The task slot number specified in the argument of the system status variables was invalid.
	Measures	Input the correct slot No.
L3890	Error message	System error (make MCODE)
	Cause	An error has occurred when creating an operation instruction. The program may have been damaged.
	Measures	Restore using the backup data. If the backup data is not available, it is necessary to create a program again.
L3900	Error message	JRC Command is disable
	Cause	The JRCEXE parameter is disabled, so it cannot be used.
	Measures	Change the JRCEXE parameter, and then execute.
L3910	Error message	Cannnot execute (JRC 0)
	Cause	JRC 0 can not execute for robot arm axis.
	Measures	Correctly set.
L3930	Error message	This command cannot be executed
	Cause	Collision detection is effective
	Measures	Repeal collision detection (execute ColChk Off)
L3940	Error message	ColChk cannot be executed
	Cause	An exclusive function is performing with ColChk
	Measures	Repeal the corresponding function
L3950	Error message	NOERR cannot be executed
	Cause	Interruption using M_ColSts is invalid
	Measures	Define interruption using M_ColSts and confirm it
L3960	Error message	This Act No. cannot be repealed
	Cause	NOERR of collision detection is performed
	Measures	Repeal this interruption after canceling NOERR
L3970	Error message	ColChk cannot be performed
	Cause	Collision detection serves as prohibition of use
	Measures	Change parameter COL into use permission
L3980	Error message	Load mode cannot be specified.
	Cause	Prec command is executed
	Measures	Repeal high accuracy mode (execute Prec Off)

Error No.	Error cause and measures	
L3982	Error message	Cannot be used (singular point)
	Cause 1	This robot does not correspond to the singular point function
	Measures 1	Check the argument of Type specification
	Cause 2	Cmp command is executed
	Measures 2	Invalidate a compliance mode (execute Cmp Off)
	Cause 3	A synchronous addition axis control is effective
	Measures 3	Invalidate a synchronous addition axis control
	Cause 4	Tracking mode is effective
	Measures 4	Invalidate a tracking mode (execute Trk Off)
	Cause 5	Pre-fetch execution is effective
	Measures 5	Invalidate a pre-fetch execution
	Cause 6	This robot is a setting of the multi mechanism
	Measures 6	Do not use the function of passage singular point
	Cause 7	ColChk On command is executed
	Measures 7	Invalidate a collision detection (execute ColChk Off)
H3983	Error message	Cannot execute Cnt movement
	Cause	A structural flag or the angle of the joint is not corresponding
	Measures	It surely positions it by Dly command etc
L3984	Error message	Cannot be passsed (singularity)
	Cause	Robot passes position which is very near the singular point
	Measures	Adjust the teaching position
H3985	Error message	Cannot be passsed (singularity)
	Cause	Robot passes position which is very near the singular point
	Measures	Adjust the teaching position

Error No.	Error cause and measures	
L3986	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Unable to enable force sense control. (Sensor)
	Cause	Unable to execute because the force sensor is not connected.
	Measures	Connect the force sensor, or delete the command that cause the error.
	Error message	Unable to enable force sense control. (Cmp command)
	Cause	It is not possible to enable force sense control function while the compliance control function is enabled.
	Measures	The force sense control function and compliance control function cannot be enabled simultaneously. If using the force sense control function, disable the compliance control function.
	Error message	Unable to execute the Cmp command. (Force sense control)
	Cause	It is not possible to enable the compliance control function while the force sense control function is enabled.
	Measures	The force sense control function and compliance control function cannot be enabled simultaneously. If using the compliance control function, disable the force sense control function.
	Error message	Unable to enable the collision detection function. (Force sense control)
	Cause	It is not possible to enable the collision detection function while force sense control function is enabled.
	Measures	The force sense control function and collision detection function cannot be enabled simultaneously. If using the collision detection function, disable the force sense control function.
	Error message	Unable to enable force sense control. (Initialization)
	Cause	It is not possible to perform initialization when starting force sense control.
	Measures	Check the parameter settings.
	Error message	This is the singular point adjacent area. (Force sense control)
	Cause	It is not possible to move the singular point adjacent area while the force sense control function is enabled.
	Measures	If moving the singular point adjacent area, disable the force sense control function.
	Error message	The force sense control status is different.
	Cause	The force sense control enabled/disabled status when resuming program operation differs from that during program operation.
	Measures	Set the force sense control enabled/disabled status to the correct status. (This occurs only once when resuming program operation.)
	Error message	This function cannot be used.
	Cause	This model is not compatible with the executed force sense control function.
	Measures	Do not use this force sense control function. Contact the maker for details on the latest compatibility status.
	Error message	Unable to change tool conversion data. (Force sense control)
	Cause	It is not possible to change tool conversion data while the force sense control function is enabled.
	Measures	If changing tool conversion data, disable the force sense control function.
	Error message	Unable to change base conversion data. (Force sense control)
	Cause	It is not possible to change base conversion data while the force sense control function is enabled.
	Measures	If changing base conversion data, disable the force sense control function.
	Error message	Unable to execute the Jrc command. (Force sense control)
	Cause	It is not possible to execute the Jrc command while the force sense control function is enabled.
	Measures	To execute the Jrc command, disable the force sense control function.
	Error message	Disable force sense control.
	Cause	JOG operation cannot be performed on your model while the force sense control function is enabled.
	Measures	Disable the force sense control function.
	Error message	Unable to perform offset cancel. (Force sense control)
	Cause	It is not possible to perform offset cancel while the force sense control function is enabled.
	Measures	If performing offset cancel, disable the force sense control function.

Error No.	Error cause and measures	
L3987	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Force sense control is disabled.
	Cause	Force sense control is disabled, and so unable to execute the command.
	Measures	Enable the force sense control function.
	Error message	Force sense control is enabled.
	Cause	It is not possible to enable force sense control again while already enabled.
	Measures	First disable the force sense control function, and then enable again.
	Error message	Unable to execute the FsGChg command.
	Cause	It is not possible to execute the FsGChg command when force control gain change is not complete.
	Measures	Review the program so that the FsGChg command is executed after force control gain change is complete.
	Error message	Unable to output log file.
	Cause	FTP processing was not properly performed.
	Measures	Check the FTP related parameter setting. Check the Ethernet cable connection. Check the FTP server settings at the computer.
	Error message	The specified log file does not exist.
	Cause	The log file for the No. specified with the FsOutLog command does not exist.
	Measures	Check whether the log file No. is incorrect.
	Error message	Unable to change the force sense status variable.
	Cause	The status variable setting is currently being used by the force sense control function and so cannot be changed.
	Measures	If changing the setting, disable the force sense control function.
	Error message	Force sense log commands executed simultaneously.
	Cause	It is not possible to create/output another file while creating (FsLog Off command) or outputting (FsOutLog command) a force sense log file.
	Measures	Process after force sense log file creation/output is complete.
	Error message	Unable to create log file.
	Cause	Unable to create a force sense log file.
	Measures	Check the amount of available record space in the robot controller.
	Error message	Unable to execute the Fsc ON command.
	Cause	It is not possible to execute the Fsc On command while changing the force control gain.
	Measures	Execute the Fsc On command after force control gain change is complete.
	Error message	Unable to perform log data related processing.
	Cause	It is not possible to execute FsLog On command file while recording force sense control log data.
	Measures	Execute the FsLog On command after log data recording is complete.
	Error message	Force sense control is enabled.
	Cause	The parameter setting is currently being used by the force sense control function and so cannot be changed.
	Measures	If changing the parameter, disable the force sense control function once.
	Error message	Unable to enable the Mo trigger.
	Cause	The Mo trigger for the FsCtrg command executed first is enabled.
	Measures	Execute after changing the control characteristics.
	Error message	Mo trigger timeout
	Cause	The Mo trigger did not turn ON within the specified time.
	Measures	Review the Mo trigger conditions and robot program.
	Error message	Unable to execute the FsCTrg command.
	Cause	The control characteristics is currently being changed. The Mo trigger for the FsCTrg command executed first is enabled.
	Measures	Execute after changing the control characteristics.
	Error message	Unable to specify the control characteristics change.
	Cause	The control characteristics change has been set with another command.
	Measures	Change the program so that the command is not executed at the same time as another command.

Error No.	Error cause and measures	
H3988	Error message	Unable to create a position command.
	Cause	It is not possible to convert linear position data to joint angle after offsetting with force control. The position after offsetting lies outside the movement range or is a singular point.
	Measures	Review the movement and, settings so that adjacents outside the movement range and singular point adjacents are avoided.
L3990	Error message	ColChk cannot be used
	Cause	This robot is a setting of the multi mechanism
	Measures	Change parameter COL into use prohibition
L3991	Error message	Mainte.Forecast cannot be used
	Cause	This robot is a setting of the multi mechanism
	Measures	Invalidate the maintenance forecast
L3995	Error message	Can't use SQ Direct and ALWENA
	Cause	SQ Direct function (parameter: IQMEM valid setting) can't be used Xrun, Xload, Xstp, Xrst, Servo and Reset Error command with always running program (parameter: ALWENA=1).
	Measures	Change ALWENA parameter to 0
L4000	Error message	System error (time out)
	Cause	There is a problem in the program's exclusive process.
	Measures	If it comes back, contact your service provider.
L4100	Error message	No. of registered file is full
	Cause	The No. of program has been exceeded.
	Measures	Delete any unnecessary programs.
L4110	Error message	Memory area is full
	Cause	The program and data have exceeded the capacity. Capacity of a program execution area is insufficient.
	Measures	Delete any unnecessary programs or data.
L4120	Error message	Too long program name
	Cause	The program name is a maximum of 12 characters with 3 extension characters.
	Measures	Set the program name to within 12 characters and 3 extension characters.
L4130	Error message	Illegal program name
	Cause	An illegal character was used in the program name.
	Measures	Only numbers and alphabetic characters can be used.
L4140	Error message	The program was not found
	Cause	The designated program was not found.
	Measures	Designate a different program, or create the designated program.
L4150	Error message	Program is faulty
	Cause	Power shutdown may have occurred during the write operation.
	Measures	The file may have been damaged. Delete the file.
L4160	Error message	Not a robot program
	Cause	The designated program is not a robot program.
	Measures	Designate a different program.
L4170	Error message	The program is being edited
	Cause	The program is being edited.
	Measures	Close the program being edited.
L4180	Error message	Program is running
	Cause	The program is running.
	Measures	Stop the program.
L4190	Error message	The program is selected
	Cause	The program is preparing to execute.
	Measures	Reset the program. The user base program is selected. Or the program to be executed is used with a different slot.

Error No.	Error cause and measures	
L4200	Error message	Cannot write to file
	Cause	Write operation is prohibited, or the file capacity is insufficient.
	Measures	1. Enable file writing. 2. Delete unnecessary files.
L4210	Error message	Too long statement
	Cause	The command statement length is limited to 240 characters.
	Measures	Delete the No. of characters in the command statement to within 240 characters.
L4220	Error message	Syntax error
	Cause	There is an error in the syntax of the input command statement.
	Measures	Re-input in the correct syntax after checking the contents.
L4230	Error message	The line No. does not exist
	Cause	There is no specified line number.
	Measures	Check the contents, and reinput the correct line No.
L4240	Error message	The statement is write protected
	Cause	The command statement is write protected.
	Measures	Cancel the write protection.
L4250	Error message	No more lines or variables
	Cause	Reading of lines and variables exceeding the registered ones.
	Measures	Check the programs.
L4300	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Too long variable name
	Cause	The variable name length is limited to 16 characters
	Measures	Shorten the variable name to within 16 characters
	Error message	Too long variable/function name
	Cause	The variable/function name length is limited to 32 characters.
	Measures	Shorten the variable/function name to within 32 characters.
L4310	Error message	Illegal character is used
	Cause	A character other than A to Z or 0 to 9 was used.
	Measures	Use the character which can be used.
L4320	Error message	The variable is write protected
	Cause	The variable is write protected.
	Measures	1. Use a writable variable. 2. Cancel the write protection.
L4330	Error message	The variable is read protected
	Cause	The variable is read protected.
	Measures	1. Use a writable variable. 2. Cancel the write protection.
L4340	Error message	The variable is not defined
	Cause	The variable has not been defined.
	Measures	Define the variable.
L4341	Error message	The user external variable is not defined.
	Cause	The user external variable has not been defined
	Measures	Define the user external variable
L4350	Error message	Duplicate definition (Val.)
	Cause	Variables already defined cannot be redefined with the Dim or Def statements.
	Measures	1. Change the variable name and define. 2. Delete the defined variable.
L4360	Error message	Same variable used (65535 times)
	Cause	Example: 1 P1=P1+P2 references P1 twice and P2 once.
	Measures	Change the program to reduce the No. of times the same variable is used.

Error No.	Error cause and measures	
L4370	Error message	Error in the array element
	Cause	1. The array elements exceeds the define range. 2. The specified variable is not an array.
	Measures	1. Correct the number of array elements within one to the maximum elements. 2. Do not specify array elements.
L4380	Error message	Cannot delete variables (used)
	Cause	Variables used in a command statement cannot be deleted.
	Measures	Delete the command statement using the variable.
L4390	Error message	Variable type combination error
	Cause	The type of the user-defined external variable is different.
	Measures	Match the variable types.
L4400	Error message	Program is faulty.
	Cause	The content of the program is abnormal.
	Measures	Delete the program.
L4420	Error message	Line No exceeds 32767
	Cause	The new line No. or line gap is large.
	Measures	Do not use line No. exceeding 32767.
L4430	Error message	Not found the string searched
	Cause	The character string searched for was not found.
	Measures	Check the program.
L4440	Error message	Duplicate definition (label)
	Cause	A label already defined cannot be redefined.
	Measures	1. Change the label name. 2. Delete the defined label line.
L4460	Error message	Argument value range over
	Cause	Argument value range over.
	Measures	Confirm the argument range, and correct the value.
L4470	Error message	Line No can't be used
	Cause	Line No can't be used
	Measures	Please use the label
L4800	Error message	System error (System base prog)
	Cause	The base program for the system could not be opened. Or, the system base program name was not specified correctly in the parameter.
	Measures	If it comes back, contact your service provider.
L4810	Error message	The global variable defined by user is not available
	Cause	The parameter "PRGUSR" is not correct.
	Measures	To use user-defined external variables, it is necessary to set the name of the program describing only variable definitions in the PRGUSR parameter.
L4811 *	Error message	The global variable redefined
	Cause	A system global variable is defined in user global.
	Measures	Correct the program.
L4820	Error message	No editing program
	Cause	The program was closed while editing. For example, the program being edited is closed when a key switch enable/disable operation is performed via the T/B during online editing on a PC.
	Measures	Try editing the program again.
L4900	Error message	System error (Prog Hndl)
	Cause	The program name used by internal processing is not normal.
	Measures	If it comes back, contact your service provider.
L4910	Error message	Robot Language is mismatched
	Cause	The parameter RLNG is mismatched. (2:MELFA-BASIC V, 3:MELFA-BASIC VI)
	Measures	Re-set RLNG or select an other program.

Error No.	Error cause and measures	
L4922	Error message	Unsupported program version
	Cause	This program was created with the new version.
	Measures	This program cannot be opened with this robot.
L4920	Error message	There is no backup data in ROM
	Cause	There is no backup data in ROM.
	Measures	Please operate after backup.
L4930	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	CavChk cannot be executed (*)
	Cause	Collision avoidance serves as prohibition of use
	Measures	Change parameter CAV into use permission
	Error message	CavChk cannot be used
	Cause	This robot does not correspond to collision avoidance
	Measures	Delete CavChk command
L4931	Error message	Cmp command cannot be executed
	Cause	Collision avoidance is effective
	Measures	Repeal collision avoidance (execute CavChk Off)
L4932	Error message	Trk command cannot be executed
	Cause	Collision avoidance is effective
	Measures	Repeal collision avoidance (execute CavChk Off)
L4933	Error message	CavChk cannot be executed
	Cause	Cmp command is executed
	Measures	Repeal compliance mode (execute Cnp Off)
L4934	Error message	CavChk cannot be executed
	Cause	Trk command is executed
	Measures	Repeal tracking mode (execute Trk Off)
L4935	Error message	NOERR cannot be executed
	Cause	Interruption using M_CavSts is invalid
	Measures	Define interruption using M_CavSts and confirm it
L4936	Error message	This ACT No. cannot be repealed
	Cause	NoErr of collision avoidance is performed
	Measures	Repeal this interruption after canceling NoErr
L4937 *	Error message	A user free area size is lack
	Cause	The size of a user free area is smaller than that of 2K
	Measures	Please set 2K or more to the size of a user free area
L4938	Error message	This robot does not correspond to collision avoidance.
	Cause	This robot does not correspond to collision avoidance.
	Measures	Parameter CAV returns disapproval.
L4939	Error message	CAV version of Robot No.n is old
	Cause	Software version of Robot No.n is old
	Measures	Interference avoidance function cannot be used because the software version of interference avoidance of the robot-n is old. Software version upgrade is required. Contact the maker.
L4941	Error message	MvSpl command cannot be executed
	Cause	Collision avoidance is effective
	Measures	The spline interpolation function and collision avoidance function cannot be enabled simultaneously. If using the spline interpolation function, disable the collision avoidance function.

Error No.	Error cause and measures	
L4950	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Error in a function syntax
	Cause	Error in the definition of a function procedure.
	Measures	Correct the definition of the function procedure.
	Error message	Error in a parameter
	Cause	Error in a parameter of the function.
	Measures	Correct the parameter of the function.
	Error message	Error in a parameter
	Cause	The elements exceeds the range.
	Measures	Correct the array element.
H5000	Error message	TB Enable key is ON
	Cause	The TB Enable key was ON in the AUTO mode
	Measures	OFF the TB Enable key, or enter the TEACH mode
L5010	Error message	AUTOENA signal is OFF
	Cause	The automatic operation possible signal is OFF.
	Measures	Turn the automatic operation possible signal ON, or enter the teach mode.
L5100	Error message	No program is selected
	Cause	A program is not selected for the designated slot.
	Measures	Select a program for the designated slot.
L5110	Error message	Continuous RUN is not possible
	Cause	A different program name has been designated.
	Measures	Designate the correct program name.
L5120	Error message	Cannot select program (not PSA)
	Cause	The specified slot is not in the program selection state.
	Measures	Reset the program.
L5130	Error message	Cannot execute servo on
	Cause	A servo OFF process is taking place.
	Measures	Wait for the servo to turn OFF before turning the servo ON.
L5140	Error message	Cannot read the file
	Cause	Reading is being carried out, or editing is being carried out.
	Measures	Close the file being edited, or read after the reading is completed.
L5150	Error message	The origin has not been set
	Cause	The origin has not been set.
	Measures	Refer to the chapters of “Setting the origin” and “Resetting the origin” which are described in the separate volumes “ROBOT ARM SETUP & MAINTENANCE”. And set the origin.
L5200 *	Error message	Parameter error (TASKMAX)
	Cause	The TASKMAX parameter setting value has been exceeded (initial value: 8, maximum value: 32).
	Measures	Reduce the number of multi tasks, or change the TASKMAX parameter.
L5210 *	Error message	Parameter error (MECHAMAX)
	Cause	The No. of multi mechanisms has exceeded the limit.
	Measures	Reduce the No. of multi mechanisms.
L5400	Error message	All robot cannot be designated
	Cause	All mechanisms cannot be specified.
	Measures	Specify an independent mechanism number.
L5410	Error message	Non-existent mode
	Cause	It has been changed to a mode other than Auto/Teach.
	Measures	Contact the maker.

Error No.	Error cause and measures	
L5420	Error message	Illegal slot No.
	Cause	A task slot number other than 1 through TASKMAX (parameter) is specified.
	Measures	Designate the correct task slot.
L5430	Error message	Illegal robot No.
	Cause	The mechanism designation is illegal.
	Measures	Designate the correct mechanism.
L5600	Error message	Cannot execute during an error
	Cause	Cannot execute during an error.
	Measures	Reset the error.
C5610	Error message	Cannot execute during STOP ON
	Cause	Cannot execute during stop signal input.
	Measures	Turn the stop signal OFF and execute.
L5620	Error message	Cannot execute during CSTOP ON
	Cause	Cannot execute during cycle stop signal input.
	Measures	Turn the cycle stop signal OFF.
L5630	Error message	Cannot execute during SRVOFF ON
	Cause	Cannot execute during servo OFF signal input.
	Measures	Turn the servo OFF signal OFF.
L5640	Error message	Cannot execute during RUN
	Cause	Cannot execute during operation.
	Measures	Stop the operation, and then execute.
L5650	Error message	Cannot execute during STOP
	Cause	Cannot execute during stop operation.
	Measures	Complete the stop, and then execute.
L5660	Error message	Edit during RUN (include ALWAYS)
	Cause	Editing cannot be performed while in operation (including continuous execution).
	Measures	Stop the program, and then execute.
C5670	Error message	Cannot execute during wait
	Cause	Cannot execute during wait
	Measures	Execute Program Reset
L5990	Error message	System error (Illegal command)
	Cause	Not exist command was executed.
	Measures	Execute the correct command.
L6010	Error message	Illegal command
	Cause	This may have occurred because data was sent before the communication line was opened via a data link, or an unregistered communication command was sent due to the mismatch between the versions of the controller and support software.
	Measures	Send after the communication line is opened. Or, match the versions.
L6011	Error message	Cannot connect to CR800
	Cause	Cannot connect to CR800
	Measures	Please connect to CPU unit
L6020	Error message	The Operation is disable
	Cause	The operation rights have not been acquired.
	Measures	Acquire the operation rights.
L6030	Error message	The editing operation is disable
	Cause	The editing operation rights have not been acquired.
	Measures	Acquire the editing operation rights.
L6040	Error message	System error (illegal device No)
	Cause	A non-registered device No. was set.
	Measures	Set a valid device.

Error No.	Error cause and measures	
C6050	Error message	The file cannot be opened
	Cause	The block file cannot be opened.
	Measures	Check the file, and designate the correct file.
C6060	Error message	The mode is not TEACH
	Cause	Carry out parameter writing in the teach mode.
	Measures	Change to the teaching mode, and then execute.
C6070	Error message	The time cannot be set
	Cause	The time setting can only be executed when the program is stopped and the servo is OFF.
	Measures	Stop the program and turn the servo OFF, and then set the time.
C6080	Error message	Com mesg is too long
	Cause	The character string of a communication text exceeded the maximum number allowed.
	Measures	Verify the number of characters in a comment in various settings such as parameters.
C6090	Error message	Can not power reset of robot
	Cause	Can not power reset of robot
	Measures	Can not power reset of robot
C6500	Error message	Not opened COM line
	Cause	Open was not executed by the program.
	Measures	Execute Open, and then send PRN.
H6530 *	Error message	COMDEV parameter is illegal
	Cause	Illegal parameter (COMDEV).
	Measures	Correct COMDEV parameter.
L6600	Error message	Signal number is out of range.
	Cause	The designated signal No. has not been defined.
	Measures	Change the signal No. to the correct No.
L6610	Error message	Cannot output (hand input)
	Cause	The hand input signal cannot be written.
	Measures	Use the correct output signal.
L6620	Error message	Cannot write (special Input)
	Cause	The input signal cannot be written into the robot dedicated area.
	Measures	Use an actual signal.
L6630	Error message	Input signal cannot be written
	Cause	This is the actual signal input mode.
	Measures	Set a pseudo-input signal.
H6640 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Illegal param (special signal)
	Cause	The parameter setting is illegal.
	Measures	Correct the changed parameter.
	Error message	Illegal setting of the dedicated signal parameter SFMODE
	Cause	The setting of the dedicated signal parameter SFMODE (safety mode output) is illegal.
	Measures	An error (output bit width of the output signal is less than 3 bits) exists in the dedicated output SFMODE setting. Correct the parameter setting.
L6641	Error message	Duplicate setting (special IN)
	Cause	The parameter setting is illegal.
	Measures	Correct the parameter of the dedicated input signal you changed.
H6642 *	Error message	STOP is fixed signal No. 0
	Cause	The parameter setting is illegal.
	Measures	Set the parameter STOP(input) to 0.

Error No.	Error cause and measures	
L6643	Error message	Illegal parameter (special Sig.)
	Cause	The parameter setting is illegal.
	Measures	Make the ending number larger than the starting number.
L6650 *	Error message	Duplicate setting (special OUT)
	Cause	The parameter setting is illegal.
	Measures	Correct the parameters.
L6651 *	Error message	Duplicate setting (HANDTYPE)
	Cause	The parameter setting is illegal.
	Measures	Correct the parameter.
L6660	Error message	Cannot output (SPECIAL OUT)
	Cause	The program setting is illegal.
	Measures	Correct the program.
L6670	Error message	Illegal OUT reset pattern
	Cause	The parameters are not defined in sets of 8 characters.
	Measures	Correct the parameters.
L6800 *	Error message	"Cancel pseudo-input mode, PW OFF"
	Cause	If pseudo input was canceled, it is necessary to turn the power ON again in order to prevent the erroneous operation of the robot by external input signals.
	Measures	Turn the power OFF and then ON once. It switches to an external input signal.
C6900	Error message	Pseudo-input signal mode
	Cause	Set with the parameters.
	Measures	To set a real signal, reset the parameter and then turn the power ON again.
C7000	Error message	Copy source file was not found
	Cause	The copy source file was not found.
	Measures	Input the correct file name.
C7010	Error message	Delete target file was not found
	Cause	The delete target file was not found.
	Measures	Input the correct file name.
C7020	Error message	Rename target file was not found
	Cause	The rename target file was not found.
	Measures	Input the correct file name.
H7030 *	Error message	System error (param size over)
	Cause	The change capacity is too large.
	Measures	Contact the maker.
C7040	Error message	Parameter changes prohibited
	Cause	Changing this parameter is prohibited as it is a dangerous parameter.
	Measures	Contact the maker.
H7050	Error message	File is illegal
	Cause	This file is damaged.
	Measures	Contact to the dealer.
H7060 *	Error message	System error (RAM area full)
	Cause	The capacity has been exceeded.
	Measures	Contact the maker.
C7070	Error message	Memory area is full
	Cause	The program and data have already exceeded the capacity.
	Measures	Delete any unnecessary programs or data.
L7071	Error message	Not enough memory area for CTN
	Cause	Continue function needs more than 100Kbytes memory area.
	Measures	Delete any unnecessary programs.

Error No.	Error cause and measures	
C7080	Error message	Can not read parameter
	Cause	Non-existent parameter or illegal password.
	Measures	1. Input the correct parameter name. 2. Input the correct password
C7081	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Can not write parameter
	Cause	Non-existent parameter or illegal password.
	Measures	1. Input the correct parameter name. 2. Input the correct password
	Error message	Illegal CRC during parameter writing
	Cause	The parameter CRC value is illegal.
	Measures	An error is detected in the CRC check in the writing process of a parameter related to the safety monitoring function. Check the communication environment between the robot controller and the personal computer, and perform the writing operation again.
C7090	Error message	Parameter comment illgal
	Cause	The number of parameter comment characters exceeded the limit.
	Measures	Operation will not be affected, so continue the operation.
L7099	Error message	Can not read parameter"xxx". "xxx": Parameter name.
	Cause	The parameter doesn't exist.
	Measures	Please confirm the parameter name.
H7300 *	Error message	Loading the parameter file.
	Cause	It is necessary to turn the power ON again to reflect the parameters.
	Measures	Turn the power OFF and ON once.
C7310	Error message	Changed variables weren't saved
	Cause	The power was OFF during program execution.
	Measures	Don't turn the power OFF during program execution.
L7311	Error message	The power was OFF during file saving
	Cause	The power was OFF during file saving.
	Measures	Don't turn the power OFF during file saving.
L7332	Error message	Can't change to RAM mode in CTN mode
	Cause	Can't change to RAM mode in continue mode.
	Measures	Please change to RAM mode and perform again.
L7340	Error message	Contine function cannot be used in DRAM mode
	Cause	Contine function cannot be used in DRAM mode
	Measures	Please change to RAM mode and perform again
L7341	Error message	Can't change to DRAM mode in CTN mode
	Cause	Can't change to DRAM mode in CTN mode
	Measures	Please change to RAM mode and perform again
L7342	Error message	Global extension cannot be used in DRAM mode
	Cause	Global extension cannot be used in DRAM mode
	Measures	Please change to RAM mode and perform again
L7343	Error message	Can't change to DRAM mode in PRGGBL mode
	Cause	Can't change to DRAM mode in PRGGBL mode
	Measures	Please change to RAM mode and perform again
L7370	Error message	The password is 8 to 32 char.
	Cause	The password is 8 to 32 characters
	Measures	Please input the password 8 to 32 characters

Error No.	Error cause and measures	
L7371	Error message	Only alphanumeric character
	Cause	It input it excluding the alphanumeric character
	Measures	Please input the password in the alphanumeric character
L7372	Error message	The password is unmatch
	Cause	The password is unmatch
	Measures	Please input the password again
L7373	Error message	Password lock to the programs
	Cause	Password lock to the programs
	Measures	Please release password
L7374	Error message	Password lock to the parameter
	Cause	Password lock to the parameter
	Measures	Please release password
L7375	Error message	Password lock to the files
	Cause	Password lock to the files
	Measures	Please release password
L7378	Error message	Change password
	Cause	The password has not been changed from the initial value.
	Measures	The password to change functional safety related parameters has not been changed from the initial value. Change the password to new one, and perform parameter setting. The factory default password is “MELFASafetyPSWD”.
C7450	Error message	No battery voltage (robot CPU)
	Cause	The battery is spent.
	Measures	The battery voltage of the robot CPU dropped to 2.5V or less. Replace the battery, and then restore the robot program and parameters. For more information about the replacement procedure, refer to the separate volumes, ″Controller setup, basic operation, and maintenance″.
C7451	Error message	Battery voltage low (robot CPU)
	Cause	The battery will be spent soon.
	Measures	The battery voltage of the robot CPU dropped to 2.7V or less. Back up the robot program and the parameters, and replace the battery immediately. Otherwise, they may be lost due to low voltage. For more information about the replacement procedure, refer to the separate volumes, ″Controller setup, basic operation, and maintenance″.
C7500	Error message	No battery voltage (robot)
	Cause	The battery is spent.
	Measures	Replace the battery and set the origin. For more information about the replacement procedure, refer to the separate volumes, ″Instruction Manual/Robot Arm Setup to Maintenance″.
C7510	Error message	Battery voltage low (robot)
	Cause	The battery will be spent soon.
	Measures	Replace the battery. For more information about the replacement procedure, refer to the separate volumes, ″Instruction Manual/Robot Arm Setup to Maintenance″.
C753n (n indicates the axis number (1 to 6).)	Error message	Replenishment time of grease
	Cause	It seems that grease reached longevity
	Measures	Replenish grease
C754n (n indicates the axis number (1 to 6).)	Error message	Exchange time of the belt
	Cause	It seems that the belt reached longevity
	Measures	Execute the check and the exchange of the belt
H7600 *	Error message	Mechanism number of additional axis is illegal.
	Cause	The value of AXMENO (mechanism No. used) parameter is illegal.
	Measures	On the mechanical additional axis, change ″0″ to a value which is smaller than set to AXUNUM (number of mechanisms used).

Error No.	Error cause and measures	
H7601 *	Error message	Axis number of additional axis is illegal.
	Cause	The value of AXJNO (setting axis No.) is illegal.
	Measures	Change the value of this parameter from 1 to 3. Change the value of this parameter to a order from 1.
H7602 *	Error message	Axis numbers of additional axis are overlap.
	Cause	As the parameter value of AXJNO (setting axis No.), the same value is set at two or more elements.
	Measures	Change the values of the element Nos. which set the same values at this pa-rameter AXMENO, to all different values.
H7603 *	Error message	Unit of additional axis is illegal.
	Cause	The value of AXUNT (unit axis) parameter is illegal.
	Measures	Change the values of all elements of this parameter to "0" or "1".
H7604 *	Error message	Acceleration time of additional axis is illegal.
	Cause	The value of AXACC (acceleration time) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive real numbers.
H7605 *	Error message	Deceleration time of additional axis is illegal.
	Cause	The value of AXDEC (deceleration time) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive real numbers.
H7606 *	Error message	Gear ratio numerator of additional axis is illegal.
	Cause	The value of AXGRTN (total speed ratio numerator) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive integers.
H7607 *	Error message	Gear ratio denominator of additional axis is illegal.
	Cause	The value of AXGRTD (total speed ratio denominator) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive integers.
H7609 *	Error message	Motor rated speed of additional axis is illegal.
	Cause	The value of AXMREV (rated speed) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive integers.
H7610 *	Error message	Motor maximum speed of additional axis is illegal.
	Cause	The value of AXJMX (maximum speed) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive integers.
H7611 *	Error message	Encoder pulse of additional axis is illegal.
	Cause	The value of AXENCR (encoder resolution) parameter is illegal.
	Measures	Change the values of all elements of this parameter to positive integers.
H7612 *	Error message	JOG smoothening time constant of additional axis is illegal.
	Cause	The value of AXJOGTS (JOG smoothening time constant) parameter is illegal.
	Measures	Change the values of all elements of this parameter to 0 or positive real number.
H7613 *	Error message	Turn OFF the power supply once, and turn ON it again.
	Cause	It is necessary to turn OFF the power supply once.
	Measures	Turn OFF the power supply of the controller, and turn it ON again.
H7650	Error message	The force sensor quantity setting is incorrect.
	Cause	Only 1 force sensor can be used for a single robot.
	Measures	Check the parameter (AXJNO, AXMENO) settings to see whether multiple force sensors have been set.
H7651 *	Error message	Force sense I/F unit initialization error
	Cause	The force sense I/F unit was not recognized, and therefore it was not possible to successfully complete initialization.
	Measures	Check the force sense I/F unit wiring and whether the power supply is ON.
H7652 *	Error message	Force sense I/F unit revision illegal
	Cause	This force sense I/F unit revision is not supported.
	Measures	Contact the maker.

Error No.	Error cause and measures	
H766n (n indicates the sensor axis.)	Error message	The force sensor data exceeded the tolerance value.
	Cause	The force acting on the force sensor exceeded the set tolerance value.
	Measures	Check whether too large a force is acting on the force sensor. Check whether an appropriate value has been set for parameter FSLMTMX. Refer to the section "Force Sensor Tolerance" in separate manual, "Instruction Manual/Force Sense Function" for details on the error recovery method.
H7700 *	Error message	CC-Link card is illegal (Error Code).
	Cause	CC-Link card is illegal.
	Measures	Please exchange the CC-Link card. When it comes back, contact to the dealer.
H7710 *	Error message	Cannot set a CC-Link master station.
	Cause	A master station is already set by the rotary switch.
	Measures	Set the rotary switch to other than 0.
H7720 *	Error message	Two CC-Link interface cards are mounted.
	Cause	Mount one card in slot 2.
	Measures	It is not allowed to install two cards. Install only one card.
L7730	Error message	CC-Link data link error (local station connection error)
	Cause	There is a line error or the master station's parameter settings are invalid.
	Measures	Review the line and parameters.
L7750	Error message	A (CC-Link) cable is not connected or parameters do not match.
	Cause	A cable is not connected or parameters do not match.
	Measures	Reset the power and start again.
H7760 *	Error message	CC-Link initialization error
	Cause	The master station's parameters do not match.
	Measures	Correct the parameters, and then start again.
L7780	Error message	A CC-Link register number is outside the range.
	Cause	A register number entered is outside the allowable range.
	Measures	Enter the correct value.
L7781	Error message	A signal number for CC-Link was specified.
	Cause	A signal number for CC-Link was specified.
	Measures	Install a CC-Link interface card.
H7810	Error message	MONPORT/SLMPPORT parameter error
	Cause	The element of MONPORT(1/2) and SLMPPORT overlap.
	Measures	Please set not to overlap to another port number.
H7820	Error message	Mxt Scommand timeout
	Cause	The setting time of the MXTTOUT parameter has been exceeded.
	Measures	Make the MXTTOUT parameter value larger.
H7830	Error message	"Ethernet card is not installed, or command disable"
	Cause	Ethernet card is not installed, or command disable.
	Measures	Install Ethernet interface card.
H7840	Error message	Mxt command Illegal received data
	Cause	The command argument and the data type do not match.
	Measures	Check the command and the data you are sending.
H7860	Error message	SLMP error
	Cause	An error related to SLMP has occurred.
	Measures	Please confirm the content by a detailed number of the error.

Error No.	Error cause and measures	
H7861	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	SLMP parameter error
	Cause	Illegal parameter.
	Measures	Correct parameter.
	Error message	SLMP SLMPPORT parameter error
	Cause	Overlap with server port number of NETPORT.
	Measures	Please set not to overlap to another port number.
	Error message	SLMP SLMPPORT parameter error
	Cause	Overlap with element of NETPORT(1) or MONPORT(1/2).
	Measures	Please set not to overlap to another port number.
L8300	Error message	There are a lot of GETPOS define
	Cause	GETPOS function are up to 8.
	Measures	Please use the same Act No. or reset an unnecessary program.
L8310	Error message	GETPOS undefined
	Cause	GETPOS undefined.
	Measures	Please define GETPOS.
H8320	Error message	System Error (GETPOS)
	Cause	Internal data of GETPOS is illegal.
	Measures	Turn the power OFF and ON once.
H8400	Error message	CTN data is illegal (Prec)
	Cause	CTN data is illegal. It changed to Prec Off.
	Measures	Try to set be a Prec mode again.
L8600	Error message	The vision is a unconnection
	Cause	The vision is a unconnection
	Measures	Please check Com No. and Parameter
L8601	Error message	The vision can't be logged on
	Cause	The parameter of the user-name or the password is abnormal
	Measures	Please set the parameter correctly
L8602	Error message	Password is abnormal
	Cause	The password of the user-name is not corresponding
	Measures	Please set the password correctly
L8603	Error message	Parameter is abnormal
	Cause	The parameter of the user-name or the password is abnormal
	Measures	Please set the parameter correctly
L8610	Error message	The communication is abnormal
	Cause	The communication was cut
	Measures	Please check the communications cable
L8620	Error message	The vision number is abnormal
	Cause	The number is not designated by NVOpen
	Measures	Please check NVOpen command
L8621	Error message	Vision program name is abnormal
	Cause	The vision program name has exceeded 15 characters
	Measures	Please make the program name within 15 characters
L8622	Error message	There is no vision program
	Cause	There is no specified vision program
	Measures	Please check the vision program name

Error No.	Error cause and measures	
L8623	Error message	SKIP number is already used
	Cause	SKIP number is already used
	Measures	Please confirm the SKIP number
L8630	Error message	The recognition cell is illegal
	Cause	There is no value on the cell
	Measures	Please check the recognition cell
L8631	Error message	The cell is outside the range
	Cause	The range of the cell is exceeded
	Measures	Please check the range of the cell
L8632	Error message	The vision is a time-out
	Cause	There is no response from the vision
	Measures	Please check the time-out time
L8633	Error message	The vision is a time-out (NVTRG)
	Cause	There is no response from the vision (NVTRG)
	Measures	Please decrease the load of the network
L8634	Error message	There is a comma within the range of the cell
	Cause	There is a comma within the range of the cell
	Measures	Please check the range of the cell
L8635	Error message	There is no comma within the range of the cell
	Cause	There is no comma within the range of the cell
	Measures	Please check the range of the cell
L8636	Error message	Vision Tag name is abnormal
	Cause	There is no specified vision tag in the vision program
	Measures	Please correct the vision tag name
L8640	Error message	The image trigger is abnormal
	Cause	The trigger setting of the vision sensor is abnormal
	Measures	Please check the trigger setting
L8650	Error message	Please make the vision online
	Cause	The vision sensor is off-line
	Measures	Please make vision sensor online
L8660	Error message	There is no authority
	Cause	The authority of the user-name is not a full access
	Measures	Please check user name
L8670	Error message	Can not restart
	Cause	It started without resetting it
	Measures	Please reset the program
L8680	Error message	Unknown code was received (code no.)
	Cause	Status code of unregistration was received from vision sensor. (Status cord which is not registered with robot controller was received.)
	Measures	The status code received from vision sensor is displayed to the code no.. When the code no. is not the numerical value, there is a possibility which received abnormal data. Please check the communication state, the communication setting, and the command argument.
L8700 (CR800-R/ CR800-Q only)	Error message	Multi CPU system self-check error
	Cause	Illegal parameter of multi CPU system and CPU is abnormal,etc.
	Measures	Confirm details and measures of the error with GX Developer etc.
L8710	Error message	PC I/O module parameter(m) error. (m): Target parameter numbers 1-4. (correspond to QXYUNIT 1-4)
	Cause	An illegal setting of the PC I/O module was detected.
	Measures	Please check the value of parameter.

Error No.	Error cause and measures	
L8720	Error message	Fuse is blown.(PC I/O module (m)). (m): Target parameter numbers 1-4. (correspond to QXYUNIT 1-4)
	Cause	Fuse is blown.(PC I/O module)
	Measures	Please refer to the manual of PC I/O module.
L8730	Error message	I/O module (m) remove error. (m): Target parameter numbers 1-4. (correspond to QXYUNIT 1-4)
	Cause	PC I/O module was removed.
	Measures	Please mount PC I/O module.
H8740	Error message	CR800 init communication error
	Cause	Initial communication with the CR800 was not possible
	Measures	Check the communication cable connection
H8741	Error message	CPU unit init communication error
	Cause	Initial communication with the CPU unit was not possible
	Measures	Check the communication cable connection
H8751	Error message	CR800 communication error
	Cause	Communication from the CR800 was cut off
	Measures	Check the communication cable connection
H8752	Error message	CPU unit communication error
	Cause	Communication from the CPU unit was cut off
	Measures	Check the communication cable connection
H8760	Error message	PLC CPU parameters are not set
	Cause	Fixed Scan Communication Area is not set in the PLC CPU
	Measures	Please set the Fixed Scan Communication Area in the PLC CPU
H8761	Error message	CPU number setting is illegal
	Cause	The setting of CPU number is different
	Measures	Please check PLC's multi CPU setting and QMLTCPUN parameter
H8762 *	Error message	Multi CPU parameter setting is illegal
	Cause	The multi CPU parameter setting is incorrect.
	Measures	Synchronization processing in the robot CPU was not properly performed due to the incorrect multi CPU parameter setting. Please check and correct the value of the multi CPU parameter setting, and then turn the power OFF and ON once.
H8770	Error message	CR800 communication error
	Cause	Communication error was detected from CR800
	Measures	Check the communication cable connection
H8780	Error message	CPU unit communication error
	Cause	Communication error was detected from the CPU unit
	Measures	Check the communication cable connection
H8790	Error message	CR800 error reset error
	Cause	Can not reset CR800 error
	Measures	Check the communication cable connection

Error No.	Error cause and measures	
H8800 *	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	ASIC communication error
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.
	Error message	ASIC CRC error
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.
	Error message	ASIC short frame error
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.
	Error message	ASIC overrunning error
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.
	Error message	ASIC connector connection error
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.
	Error message	ASIC 0byte transmission
	Cause	A failure was detected in the communication within the controller. The internal temperature of the controller may be high. For the CR800-R/CR800-Q controller, the following conditions may also be the cause. ・The power on timing may be incorrect. ・The cable between the controller and the robot CPU may have a fault.
	Measures	Confirm that the controller filter has no clogging and the ambient temperature is 40 ℃ or less. If the error persists after taking above measures, contact the manufacturer. For the CR800-R/CR800-Q controller, turn on the switch of the controller, then of the robot CPU. Check the cable connection between the robot CPU and the controller.

Error No.	Error cause and measures	
H8810	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Communication error
	Cause	Error occurred by the data transmission to the servo
	Measures	Check the communication cable connection. Check that there is not a source of noise.
	Error message	Illegal ID
	Cause	ID of the com. data with the servo is not corresponding
	Measures	Turn the power OFF and ON once
H8820	Error message	Communication timeout
	Cause	The communication with the servo became a timeout
	Measures	Turn the power OFF and ON once
H8830	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Amplifier unconnection
	Cause	The servo amplifier cannot be detected
	Measures	Check the cable connection and the power supply of amp.
	Error message	Servo axis setting error
	Cause	The axis setting is not correct
	Measures	Check the servo axis settings (parameters, rotary switches)
	Error message	Uncorrespondence amplifier
	Cause	Uncorrespondence amplifier is connected
	Measures	Remove the uncorrespondence amplifier
	Error message	Uncorrespondence protocol
	Cause	The communication protocol which does not correspond
	Measures	Remove the uncorrespondence amplifier
	Error message	Illegal motor ID
	Cause	Uncorrespondence motor is connected
	Measures	Remove the uncorrespondence motor
	Error message	Parameter setting failure
	Cause	A set value of the servo parameter is illegal
	Measures	Change to a correct value
	Error message	Illegal amp. number
	Cause	The number of connected amplifiers is not suitable
	Measures	Check the number of connected amplifiers
H887n	One of the errors below is detected. Please take measures corresponding to an error message.	
	Error message	Transient processing error
	Cause	Illegal demand data was transmitted
	Measures	Turn the power OFF and ON once. If it comes back, contact to your service provider.
	Error message	Transient processing timeout
	Cause	There is no response to the demand from servo
	Measures	Turn the power OFF and ON once. If it comes back, contact to your service provider.
H8920	Error message	Sensor I/F unit error (**) Note) "**" is substituted with the "sensor I/F unit error No." (2 hexadecimal digits)
	Cause	An error occurred at the force sensor interface unit.
	Measures	See Page 73, "Table 2-1: Force sense interface unit errors" based on the error No. in the Err. Message.
C8921	Error message	Sensor I/F unit warning (**) Note) "**" is substituted with the "sensor I/F unit warning No." (2 hexadecimal digits)
	Cause	A warning occurred at the force sensor interface unit.
	Measures	See Page 73, "Table 2-1: Force sense interface unit errors" based on the error No. in the Err. Message.

Error No.	Error cause and measures	
H9000 ～ H9099	Error message	User High level error
	Cause	A high-level alarm was issued from the robot program.
	Measures	Check the program.
L9100 ～ L9199	Error message	User Low level error
	Cause	A low-level alarm was issued from the robot program.
	Measures	Check the program.
C9200 ～ C9299	Error message	User Caution level error
	Cause	A warning was issued from the robot program.
	Measures	Check the program.

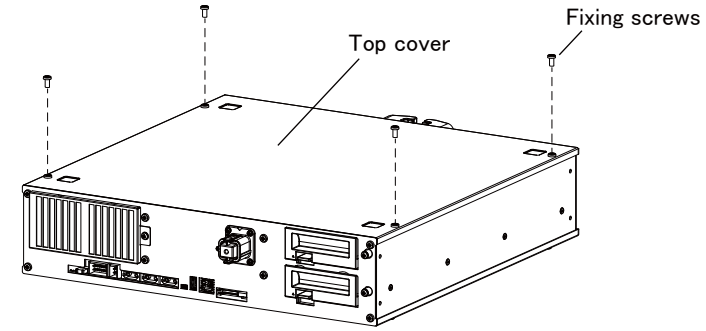
2 Appendix

2.1 Place where fuse replacement is required

2.1.1 Place where a brake fuse replacement is required.

Remove the top cover of the controller and replace the fuse (LM20) of the servo CPU board (DQ865n).

Remove the four top cover fixing screws (M4 x 8), and remove the top cover.



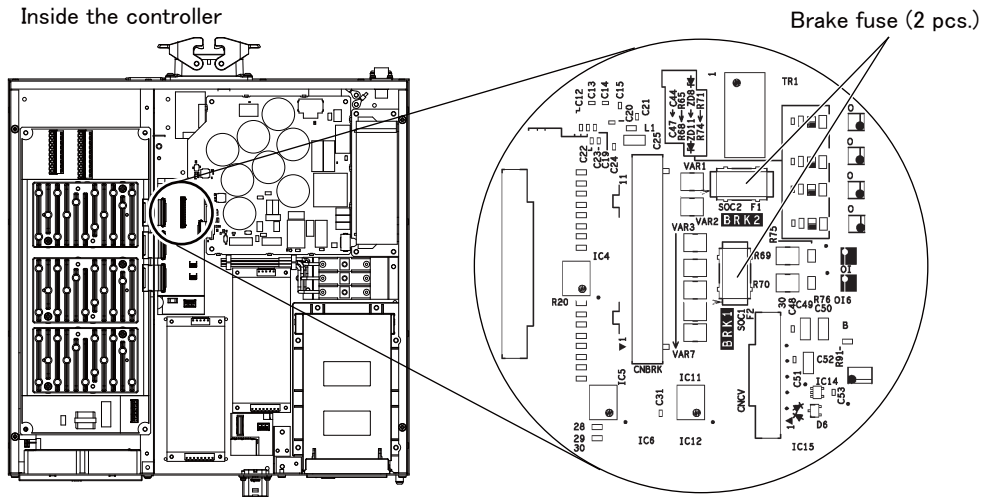


Fig.2-1 : Brake fuse exchange place

2.1.2 Place where hand fuse replacement is required

Replace the fuse (LM16) of HAND FUSE at the front of the controller.

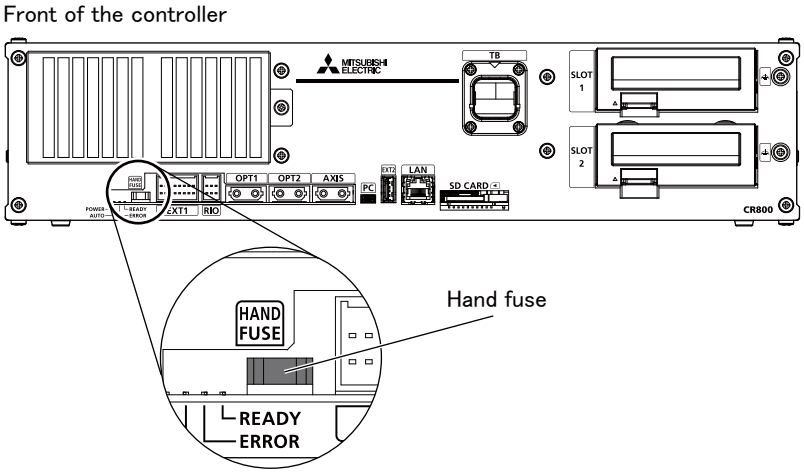


Fig.2-2 : Hand fuse exchange place

2.1.3 Place where fuse replacement is required.

Remove the bottom cover of the controller and replace the fuse (LM40) of the CPU board (DQ171n).

Remove the four bottom cover fixing screws (M4 x 8), and remove the base plate.

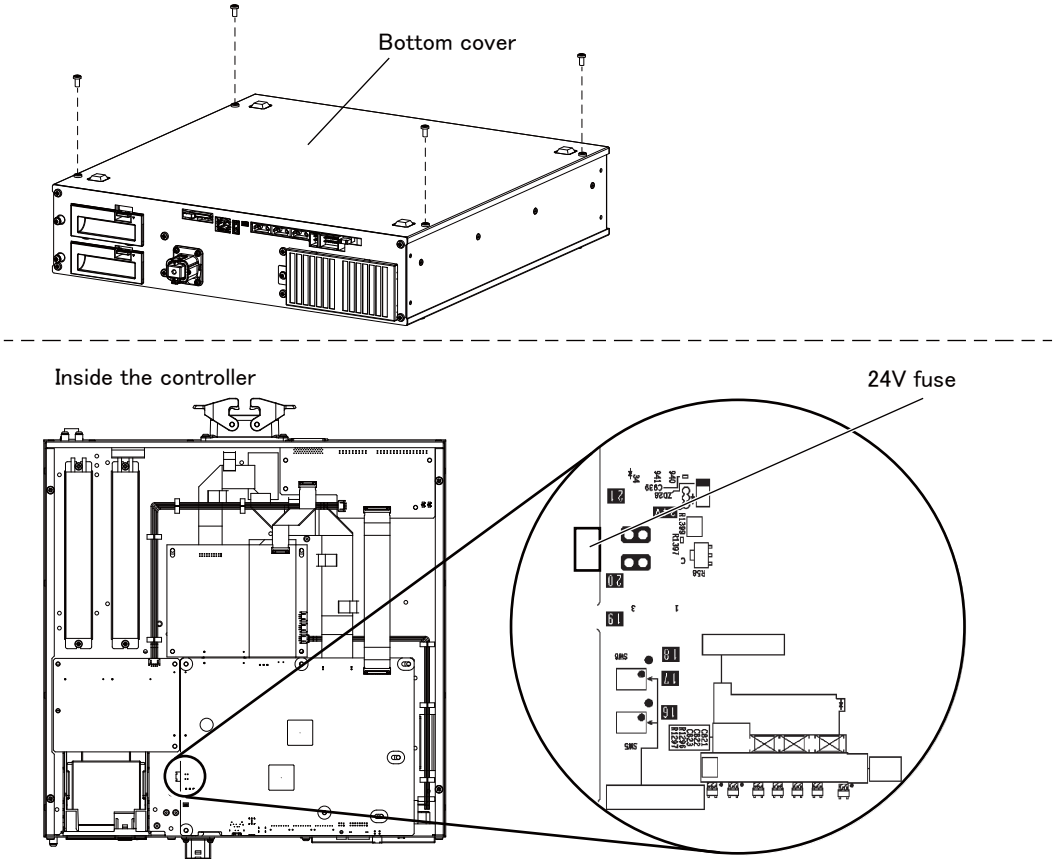
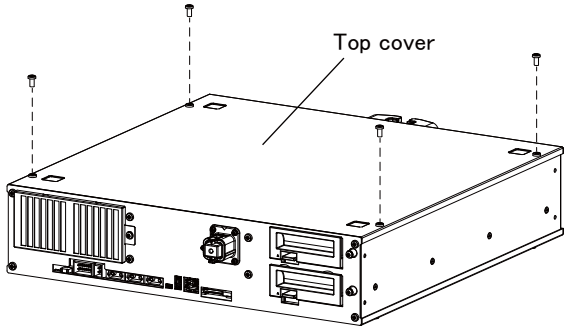


Fig.2-3 : Fuse exchange place

2.1.4 Place where a converter fuse replacement is required

Remove the top cover of the controller and replace the fuse of the converter board (DQ801n).

Remove the four top cover fixing screws (M4 x 8), and remove the top cover.



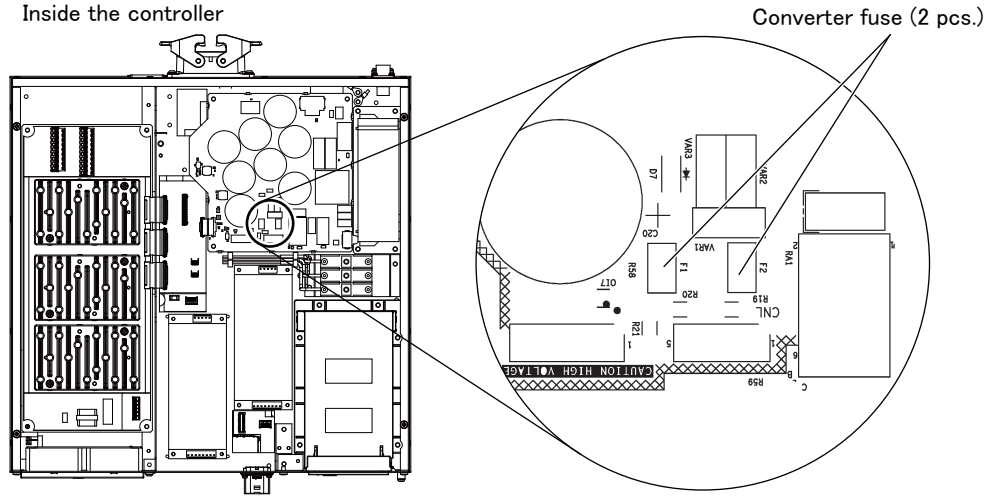


Fig.2-4 : Converter fuse exchange place

2.2 Fan installation place of robot controller.

The installation place of the fan of the robot controller is shown in the following.

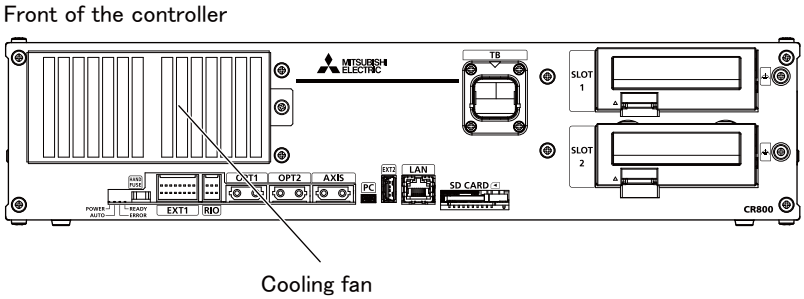


Fig.2-5 : Fan installation place

2.3 Troubles and measures

No.	Issue	
1	An unusual noise can be heard (from the robot arm).	
	Cause	・Unusual noises can be caused by friction between movable parts, deformed parts, foreign matters getting inside, and defects in belt tension. An unusual noise is also made from the fan and the cover bolts became loosened.
	Measures	・Identify the particular point from where the unusual noise is being made, and where necessary treat the problem by replacing parts, etc. ・ When operating at slow speeds, a rattling sound can be heard from the brakes, but this is not unusual. However, if the noise gets louder as the speed gets higher, contact the manufacturer.
2	An unusual noise can be heard (from RC)	
	Cause	・ A noise is made as the internally mounted fan and its casing resonate. ・ A noise is made by the fan catching internal wiring.
	Measures	・ Check the state of the cover fastening. ・ Check to see whether or not the fan is interfering with cables, etc. If no improvement is made after carrying out the above measures, please contact the manufacturer.
3	The tip of the robot is vibrating.	
	Cause	・ The load conditions exceed the specified values (mass, inertia) ・ The load condition settings are not suitable (HNDDAT and WRKDAT settings are not consistent with the load) ・ The tip is being used near a singular point. ・ Acceleration or deceleration is too sudden. ・ The installation stand is not rigid enough. ・ Vibrations from nearby equipment are being felt.
	Measures	・Check the load conditions and the parameter settings. Please also try to adjust the acceleration/deceleration time, the speed, and the teaching point where necessary. ・ Check the condition of the unit's installation (including external vibrations). If no improvement is made after carrying out the above measures, please contact the manufacturer.
4	The servo-on cannot be performed	
	Cause	・ A servo OFF signal (SRVOFF) is being input from an external source. ・ The operating right is not given. (IOENA parameters settings/exclusive input and output). ・ An error is occurring. ・ TB operation error (enabling switch, activate/deactivate)
	Measures	・ Check for a servo OFF signal and check the operation rights. ・ Check how to operate the TB. Check that the 3 position enabling switch is set to the middle position. ・ If an error message is occurring please cancel it. If no improvement is made after carrying out the above measures, please contact the manufacturer.
5	In automatic drive mode the system does not start up even if a startup signal is input. The robot stops during automatic drive.	
	Cause	・ A stop signal (STOP/STOP2/SKIP) is being input from an external source. ・ The operating right is not given. (IOENA parameters settings/exclusive input and output). ・ An error is occurring. ・ A start signal (START) has not been input. ・ The slot attribute settings are not set to START. ・ The program's operation commands have not been executed.
	Measures	・ Check for a stop/startup signals and check the operation rights. ・ Check the attributes of the program. ・ With the robot stopped, check the status of the operation program's execution. ・ If an error message is occurring please cancel it. In no improvement is made after carrying out the above measures, please contact the manufacturer.

No.	Issue	
6	The position becomes offset during automatic drive. The position becomes offset after moving the equipment.	
	Cause	・Program error (arithmetic processing, etc.). ・Teaching operation error. ・Peripheral equipment is offset. ・There is a setting error in the origin position data. ・The origin position data has been lost (battery life, etc.). ・Origin positions are offset (the arm was moved while the power supply was OFF, etc.) ・Connecting parts have come loose. ・Belt teeth have been skipped. ・Increased backlash from the reduction gear.
	Measures	・Check the positioning data in the area where the positioning has become offset. ・Check the peripheral equipment and the robot's connecting parts (hand, robot installation section, etc.) ・Check the remaining time on the robot battery's life. ・Check the position of the origin positions. ・Check the condition of the belt. ・Check whether there is any nearby interference, etc. If no improvement is made after carrying out the above measures, please contact the manufacturer.
7	The leakage current breaker trips when the controller's primary power switch is turned ON. The leakage current breaker trips when the servo is switched on.	
	Cause	・Incorrect breaker selection (current capacity/drive purpose, etc.). ・Leakage current due to electrically conductive foreign matters (oil/moisture) getting inside. ・Earthing due to cable wear.
	Measures	・Check the specification of the breaker. ・Check whether or not electrically conductive foreign matters have got inside the robot's body or inside the controller. ・Check whether or not the cable is damaged.
8	Oil is leaking from the robot's body	
	Cause	・Deterioration in the reducer oil seal. ・Too much grease was applied. ・A grease other than the specified grease was applied. ・Excess grease left on the unit when grease was applied. ・Foreign materials have stuck to the oil seal (dust, etc.).
	Measures	・Check the amount of grease that should be applied and the name and product number of the grease. ・Take measures to make sure that dust, etc. does not adhere to joints. ・After applying grease wipe off any grease that is protruding.
9	Breaker trips/overcurrent error occurs (cutting oil gets inside the robot body)	
	Cause	・Deterioration in the packing. ・Scaling on the packing. ・Using a highly permeable cutting oil. ・An environment that exceeds the IP functionality.
	Measures	・In a mist environment, pull off and replace the packing every time the cover is removed. ・If a highly permeable cutting fluid is being used, or in an environment that exceeds the IP functionality, purge the air and take measures on the jacket. ＊Cutting oil that has got inside the arm must be thoroughly cleaned away or the components inside the arm are at risk of rotting.

No.	Issue	
10	The hand does not move properly (air hand)	
	Cause	・An air leak/break in the piping. ・The solenoid bulb has failed. ・Wire breakage on the hand signal line. ・Hand signal line short circuit ・The hand IF card has failed or has been inserted incorrectly. ・Sink/source setting error. ・Hand connecting wire error ・Insufficient air pressure. ・A foreign material has got stuck in the air hose. ・Faulty connection in the cables that run between devices.
	Measures	・Carry out a check on the air hose, joints, air pressure, and connecting wire. ・Check the condition of the hand input/output signal. ・Check whether or not the controller's sink/source settings and wiring are consistent with each other. ・Check the condition of the connections on the cables that run between devices. In the event that no improvement is made after carrying out the above measures, please contact the manufacturer.
11	The hand does not move properly The hand operation screen is not displayed on the TB (Electric operated hand)	
	Cause	・Wire breakage on the hand signal line. ・The electronic hand controller (manufactured by TAIYO Ltd.) has failed. ・Faulty connection in the cables that run between devices. ・The software version on the TB and robot controller is not compatible with the electric operated hand. ・The user is trying to use the electric operated hand on a model that is not compatible with the electric operated hand.
	Measures	・Check the connections. ・Check the condition of the hand input/output signal. ・Check the mating condition of the cables that run between devices. ・Check whether the software version is compatible or not. ・Check the model compatibility. If no improvement is made after carrying out the above measures, please contact the manufacturer.
12	The hand does not move properly The hand operation screen is not displayed on the TB (Multi hand)	
	Cause	・An air leak/break in the piping. ・The solenoid valve has failed. ・Wire breakage on the hand signal line. ・The multi hand dedicated circuit has failed. ・Insufficient air pressure. ・A foreign material has got stuck in the air hose. ・Faulty connection in the cables that run between devices. ・Wire breakage on the hand signal line. ・The software version on the TB and robot controller is not compatible with the multi hand. ・The user is trying to use the multi hand on a model that is not compatible with the multi hand.
	Measures	・A check on the air hose, joints, air pressure, and connections. ・Check the condition of the hand input/output signal. ・Check the connections. ・Check the mating condition of the cables that run between devices. ・Check whether the software version is compatible or not. ・Check the model compatibility. If no improvement is made after carrying out the above measures, please contact the manufacturer.
13	Brake is not working properly	
	Cause	[Occurs only when operating manually] ・T/B operation error. [Occurs when operating manually and automatically] ・Faulty connection of the cables that run between devices. ・Brake failure.
	Measures	[Occurs only when operating manually] ・Check the T/B operation (operation buttons, 3 position enabling switch, etc.). [Occurs when operating manually and automatically] ・Check the mating of the cables that run between devices. If no improvement is made after carrying out the above measures, please contact the manufacturer.

No.	Issue	
14	The communication fault occurs by the equipment connected to the Ethernet cable inside the robot arm.	
	Cause	・The Ethernet cable is not connected surely. ・There are the device etc. which cause the noise and the noise has applied to the Ethernet cable.
	Measures	・Confirm that the Ethernet cable is connected surely. ・If the effect by the noise can be considered, find out the noise source and remove the noise. Moreover, adds the grounding and ferrite core of the Ethernet cable if needed. Recommendation ferrite core: E04SR301334 (SEIWA ELECTRIC MFG. Co.,Ltd.)
15	The T/B does not display anything.	
	Cause	・The fuse installed in the controller may have blown out. For the fuse blowout, there may be a ground fault or short circuit in the T/B cable. ・The T/B might be out of order.
	Measures	・Investigate and correct the ground fault or short circuit portion in the wiring made by the customer. If no improvement is made after carrying out the above measures, please contact the manufacturer.

2.4 Errors involving change in specification

The specification of the following errors changes when the safety monitoring function used in the robot safety option (4F-SF002-01) is enabled.

To reset the following errors while the safety monitoring function is enabled, reset the power supply.

Error	Error message
H0039	Door Switch Signal line is faulty.
H0048	Faulty line (Mode Selector switch)
H0049	Faulty Line (T/B Enable Switch)
H0051	Wiring of the external emergency stop is abnormal.
H0061	EMG line is faulty.(O.Panel)
H0071	EMG line is faulty.(T.Box)
H0074	Faulty line (T/B Enable/Disable).
H1680	Cannot servo ON (timeout)
H1681	Unexpected servo OFF
H1682	Servo ON Timeout (Safety relay).
H1683	Servo ON Timeout. (Contactor)

2.5 Force sense interface unit errors

The errors of the force sense interface unit used in the force sense function option.

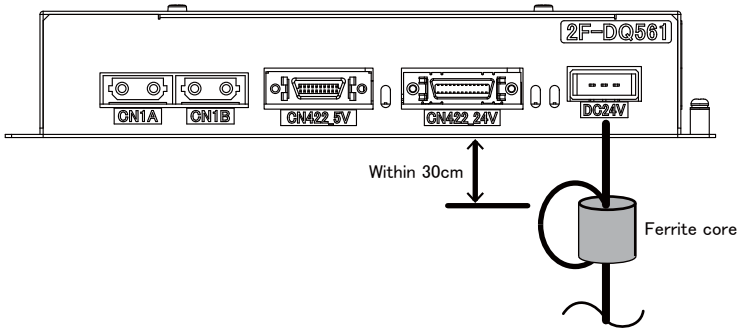
Table 2-1 : Force sense interface unit errors

Error No. (Name)	Cause	Remedy
12 (memory error)	Force sense interface unit internal part fault	Replace the unit.
13 (S/W processing error)		
21 (sensor initial communication error)	1) The force sensor connection cable is disconnected. 2) The force sensor connection cable is damaged. 3) Noise contamination occurred.	1) Connect the cable. 2) Replace the cable. 3) Perform noise countermeasures.*1)
25 (sensor communication error)		
34 (communication data error)	1) The SSCNET III cable is disconnected. 2) The SSCNET III cable end face is dirty. 3) The SSCNET III cable is damaged. 4) Noise contamination occurred.	1) Connect after turning OFF the power. 2) Wipe any dirt from the end face. 3) Replace the cable. 4) Perform noise countermeasures.*1)
36 (communication error)		
38 (communication frame error)		
39 (communication axis information error)		
37 (parameter error)	The parameter value sent from the robot controller at the power-on is incorrect.	Correct the setting value of parameter FSFLCTL.
88 (watchdog)	The robot controller does not operate normally.	1) Turn on the power supply of the robot controller again. 2) If it comes back, contact to your service provider.
E4 (parameter warning)	The input value of the parameter is incorrect.	Correct the value input in parameter FSFLCTL.

*1) Measures against noise are as follows. Take appropriate measures for the usage environment. You do not always have to take all the measures. Perform them as necessary.

1) Install a noise filter on the power supply of peripheral equipment.

2) Add a ferrite core to the 24 V power supply output cable. (Refer to the following figure.)
Install the ferrite core properly to avoid weighting the cable and connector.



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